Does an Asymmetric-Reward E-DDQN Improve Safety-Prioritized Cybersickness Detection in Virtual Reality? A Leakage-Controlled Comparison with Standard DQN and Cost-Sensitive Baselines

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***Abstract*—Reinforcement-learning (RL) detectors with asymmetric rewards have been proposed for safety-prioritized detection. We ask whether one, E-DDQN—a dueling Double Deep Q-Network whose reward penalizes a missed cybersickness event five times more than a false alarm—improves detection in virtual reality under leakage-controlled evaluation. On logged data the task is a contextual bandit, and the 5:1 reward induces exactly the 5:1 cost-sensitive Bayes rule. On the public UFFCSData gameplay dataset (35 sessions, 9,390 one-second samples), consecutive seconds are near-duplicates: random row-level splits give ROC-AUC 0.992–0.995, against 0.523 under session-grouped splits. Under a nested, session-grouped protocol fixed in advance, E-DDQN had a mean 5:1 cost of 0.663 and ROC-AUC 0.538, against 0.584 and 0.661 for a standard DQN and 0.491 and 0.688 for a cost-weighted random forest (seed-ensemble difference: Holm-adjusted $p = 0.006$ by group bootstrap, 0.083 by exact sign-flip test). With two further random search draws, specified before being run, E-DDQN never had a significantly lower cost than the standard DQN, but across draws its within-fold ROC-AUC ranged from 0.539 to 0.714: its weak primary result reflects its search draw. A post-hoc logistic regression on elapsed time alone reached ROC-AUC 0.704; no primary model ranked seconds significantly better. Every model's cost-optimal threshold raised alarms on 82–97 % of non-event seconds, and bounded telemetry perturbations changed ROC-AUC by up to 0.089. Asymmetric rewards encode a cost-sensitive operating point, not a new learning objective; RL detectors should be compared with cost-weighted and exposure-time baselines, over several search draws, under participant- or session-grouped evaluation.**

I. Introduction

Immersive virtual reality (VR) is used in clinical rehabilitation, industrial and defense training and human-performance assessment, where the headset shapes perception, balance and spatial orientation rather than merely presenting information. Cybersickness—nausea, disorientation, oculomotor strain and, in severe cases, postural instability—is a long-standing side effect of the sensory conflict that VR creates [1]. It can also be induced deliberately: an adversary who manipulates rendering parameters such as frame pacing, camera motion or field of view could make users sick or unsafe [2], [3], and attacks on the inputs of cybersickness detectors can stop a mitigation from being triggered [4]. In both cases a detector that recognizes cybersickness during immersion, so that a mitigation can be triggered, is a safety function, and its two errors are not equally costly: a missed event leaves the user exposed to harm, whereas a false alarm causes a brief, recoverable interruption.

Detectors have progressed from post-exposure questionnaires to deep temporal and multimodal models of physiological, eye-tracking and gameplay data [5]–[10]. To encode unequal error costs, several authors have cast classification as a sequential decision problem and trained deep Q-networks with class-dependent rewards [11], [12]. E-DDQN follows this line for cybersickness: a dueling Double DQN [13], [14] whose reward penalizes a missed event five times more than a false alarm, trained with class-balanced replay and soft target updates. Unlike [11], [12], it does not end an episode when an event is missed; episodes end only with the recorded session.

Two questions must be answered before such a detector can be credited with a safety benefit. The first concerns evaluation. Gameplay telemetry is logged about once per second and self-reported discomfort changes rarely within a session, so any split that places seconds of the same session or participant in both training and test data rewards recognition of the session rather than detection of cybersickness. Oversampling before splitting is a known source of such leakage [15], and published results on the gameplay dataset used here were obtained with a 70/30 split of the data for which no participant- or session-level separation is reported [16]. An earlier, unpublished version of the present study reported 96.88 % accuracy for E-DDQN under an oversampling-before-splitting protocol. The second question concerns the formulation. It is rarely made explicit what an asymmetric reward adds to classical cost-sensitive learning [17], and RL detectors are seldom compared with cost-weighted supervised models tuned and thresholded under the same rules.

We therefore address three research questions on the public UFFCSData dataset [18]:

- RQ1 What does the asymmetric reward add, formally, to cost-sensitive classification?
- RQ2 Under leakage-controlled, session-grouped evaluation, does E-DDQN outperform a standard DQN, cost-weighted supervised models and a minimal exposure-time baseline, and does any model meet a safety-prioritized operating target?

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RQ3 As a sanity check of what the detectors rely on: how sensitive are they to bounded manipulation of the telemetry channels that an adversary could influence?

The contributions are:

1) *Formal link between RL and cost-sensitive learning.* Applying known results on cost-sensitive classification and contextual bandits [17], [19], we make explicit that, on logged data, the alarm cannot change future observations, so detection is a contextual bandit, and that the 5:1 reward induces exactly the Bayes decision rule $P(y = 1 \mid s) \geq 1/6$ of 5:1 cost-sensitive classification; we also show how error-triggered episode termination, used in earlier DQN classifiers, departs from this setting, and that class-balanced replay shifts the operating point actually learned (Section III-D).
2) *A matched, leakage-controlled benchmark.* We compare E-DDQN with a standard DQN, a contextual-bandit variant and four cost-weighted supervised models under a nested, session-grouped protocol fixed before testing, with the same search procedure and the same number of configurations (16 per model; 12 for logistic regression; the search spaces differ greatly in size), the same checkpoint and threshold rules and paired statistics; two exposure-time baselines were added post hoc under the same protocol (Sections III-B–III-G). Session grouping is required on these data: random row-level splits, with or without oversampling, give ROC-AUC 0.992–0.995, against 0.523 under session-grouped splits, which quantifies a known pitfall [15], [20] on this dataset; a reconstruction of the earlier version's protocol yields similarly inflated scores (Section V-A).
3) *Findings.* E-DDQN does not have a significantly lower 5:1 cost than the standard DQN, with its primary search draw or with two further draws specified before they were run, and in ranking each agent was significantly better in one draw; with its primary draw its seed-ensemble cost is higher than that of a cost-weighted random forest in every outer fold (marginally in one; significant under the group bootstrap but not under an exact sign-flip test after Holm adjustment); in a post-hoc analysis, no primary model ranks seconds significantly better than elapsed time alone; and no model combines high recall with a controlled false-alarm rate (Section V).
4) *Diagnosis and stress tests.* E-DDQN's weak primary result is largely a consequence of its random search draw, which was dominated by configurations that use the questionnaire answers: across three draws its within-fold ROC-AUC ranged from 0.539, the lowest of the primary models, to 0.714, close to the highest, and restricted to exposure features it ranked seconds about as well as the standard DQN (Section V-G); bounded telemetry perturbations change 87 % of the detectors' scenario-wise ROC-AUC values by less than 0.03 (at most 0.089), which is not evidence of robustness, whereas an adaptive adversary still changes decisions at fixed thresholds (Section V-I).
5) *Reproducibility.* Code, the fixed protocol and per-sample predictions are released so that every number in the paper can be regenerated from the raw data.

Section II reviews related work, Section III presents the methods, Section IV the setup, Section V the results and Section VI the discussion; Section VII concludes.

II. Related Work

A. Cybersickness Detection and Its Evaluation

Questionnaires such as the Simulator Sickness Questionnaire [21] measure discomfort after exposure and cannot capture onset during immersion [1], which motivates sensor-driven detection. Dennison et al. [5] related physiological signals to cybersickness reports; Islam et al. detected and forecast severity from heart rate, breathing, heart-rate variability and skin conductance [6] and later fused eye- and head-tracking data [7]; Liu et al. used EEG with a CNN-ECA-LSTM network [8]; Yalcin et al. combined bidirectional LSTMs with GAN-based augmentation of consumer-grade physiological data [9]; and Jeong and Han applied transformers to multimodal time series [10]. Explainable models have been used to identify dominant features and to shrink detectors for deployment [16], [22], [23]. A systematic review notes that most detectors classify the current state rather than anticipating onset, and that evaluation protocols vary widely [24].

The gameplay data used here were collected by Porcino et al. [18], [25], who used symbolic machine learning to identify likely causes of discomfort such as rotation and acceleration [26]. The same gameplay data were used by TruVR, which reported 94.10 % accuracy with explainable boosting machines on a 70/30 split of the data [16], and in work on attacks against cybersickness detectors [4]. No participant- or session-level separation is reported for that split. If seconds of the same session fall on both sides of a split, 1-Hz samples with slowly changing labels allow a model to recognize sessions it has already seen [15], [20], [27]. We quantify this effect on the gameplay data in Section V-A; we do not re-evaluate the cited studies, whose exact pipelines are not available to us.

B. Adversarial Threats to VR Users and Detectors

VR mediates perception and balance and therefore exposes an attack surface distinct from conventional software [28], [29]. Tseng et al. showed that perceptual manipulations such as redirected walking can be repurposed to cause physical harm [2], and Cheng et al. studied user reactions to perceptual-manipulation attacks in mixed reality [30]. Odeleye et al. described frame-rate manipulation attacks that can induce VR sickness and detected them with an isolation forest [3]. Detectors themselves can be attacked: Kundu et al. crafted adversarial input perturbations that stop deep cybersickness detectors from triggering mitigation, evaluated them on two public datasets including the gameplay data used here, and proposed an XAI-guided attack detector [4]; Ahmed et al. built an open-source testbed in which gradient-based attacks (MI-FGSM, PGD, C&W) degrade deep detectors that drive a field-of-view mitigation in closed loop [31]. These works

perturb detector inputs by optimization; the stress tests in this paper instead apply bounded, physically motivated manipulations to the logged telemetry channels (Section III-H), and neither approach measures what a manipulation would do to a participant.

C. Reinforcement Learning for Imbalanced or Cost-Sensitive Detection

Deep Q-networks [32] overestimate action values, which Double DQN reduces by decoupling action selection and evaluation [13]; dueling heads separate state value from action advantage [14]. Lin et al. cast imbalanced classification as a sequential decision process solved by DQN with larger rewards for minority-class samples [11], and Yang et al. combined dueling and double Q-learning with a custom reward for multi-class imbalanced clinical prediction [12]; in both, a training episode ends when a minority-class sample is misclassified. E-DDQN applies the same ingredients to cybersickness, without this error-triggered termination. Safe-RL research shows that penalties shape how risk-averse a learned policy is [33], but the relation between asymmetric rewards and classical cost-sensitive thresholding [17]—and the known equivalence between a contextual bandit with full reward information and cost-sensitive classification [19]—is rarely made explicit for RL detectors, and RL detectors are seldom compared with cost-weighted supervised models under the same tuning and threshold rules.

D. Gap Addressed

Table I summarizes the gaps this study addresses. The common thread is that a safety benefit can only be attributed to a detector design when the evaluation controls leakage, the comparison is matched, and the decision rule is stated in terms of the costs it encodes.

III. Methods

The methodology has nine steps (Fig. 1). Steps 1–3 build the data so that no session, linkage group or preprocessing statistic is shared between training and test data: rebuild the data and define the target (Section III-A), make every split at the level of independent session groups before anything is learned (Section III-B), and compute causal features with preprocessing fitted on training data only (Section III-C). Step 4 states formally what the asymmetric reward means (Section III-D). Steps 5 and 6 define E-DDQN and its matched baselines (Sections III-E and III-F). Step 7 tunes, thresholds and compares every model with one protocol that uses validation data only and was fixed before testing (Section III-G). Steps 8 and 9 stress-test and explain the trained detectors (Sections III-H and III-I). Table II defines the main terms.

A. Step 1: Data and Target

Dataset. UFFCSData [18], [25], [26] was recorded while volunteers played a race game and a flight game in a head-mounted display. About once per second the game logged the player's speed and acceleration, three camera-rotation components (X, Y and Z; their range $[-1, 1]$ is consistent with the vector part of a unit quaternion, but the documentation does not state the parameterization), the frame rate, the region of interest, session-configuration flags and the participant's self-reported discomfort level on a 0–3 scale (none, slight, moderate, severe) based on the Simulator Sickness Questionnaire [21], [34]. Each session also stores the participant's answers to a Cybersickness Profile Questionnaire (CSPQ), completed before playing: gender, age band, VR experience, glasses, vision problems, eye dominance and flicker sensitivity. Two further profile fields, a symptoms field and posture, are constant in the curated release.

Provenance. We rebuilt every input from the original XML exports rather than from derived files. The dataset authors' curated release (35 seated sessions recorded in December 2019; 9,390 samples) matches 35 of the raw exports sample by sample and is our evaluation set. The 37 standing-posture sessions recorded in October 2019 are used *only* to debug code and calibrate training budgets; two further December sessions that the dataset authors excluded are not used.

Target. The binary target is $y_t = 1$ (*cybersickness event*) if the reported discomfort level at second t is at least 1, and $y_t = 0$ otherwise. The prevalence is 44.2 % (Flight 41.2 %, Race 48.1 %). Grades 2–3 occur in 1,450 of 3,993 Race samples but in only 41 of 5,397 Flight samples. The discomfort level is piecewise constant within a session: it changes on average 2.0 times per session, at irregular times, and the binary label 1.1 times; it never decreases in 23 of the 35 sessions, and 9 sessions never report discomfort. How and how often participants were prompted to report during play is not specified in the public documentation available to us, so consecutive labels should be read as a held self-report rather than as independent per-second measurements. We stress that y marks self-reported discomfort, *not* the presence of an attack: no session in the dataset was attacked.

B. Step 2: Unit of Independence and Nested Grouped Splits

Why grouping is needed. Consecutive seconds of a session have almost identical features and usually the same label. If seconds of one session appeared in both training and test data, a model could score well by recognizing the session rather than by detecting cybersickness; Section V-A quantifies this effect. Every split must therefore keep each session—and, as far as the data allow, each participant—on one side.

Session-linkage groups. The release contains no participant identifier, and sessions are not independent: some participants restarted after an aborted session (e.g., a 104-s session followed less than two minutes later by a full session with an identical nine-field profile). We therefore link two sessions if (a) their nine profile fields are identical, or (b) one starts within 180 s after an aborted session (fewer than 200 samples) and their profiles differ in at most two fields. The connected components of these links are the *session-linkage groups*; there are 21 of them. The rule may merge different people who share a profile, which only makes the evaluation more conservative, but it cannot detect one person who answered the questionnaire differently in non-consecutive sessions; we discuss this residual risk in Section VI-E.

TABLE I
GAPS IN THE EVALUATION OF ASYMMETRIC-REWARD CYBERSICKNESS DETECTORS AND HOW THIS STUDY ADDRESSES THEM.

Gap	Evidence	This study
Splits of 1-Hz session data without participant or session separation	70/30 data split without reported participant or session separation on the same gameplay data [16]; leakage literature [15], [20], [27]	Session-linkage groups, nested grouped evaluation, train-only preprocessing, protocol fixed before testing; effect of the split protocol reported as motivation (Section V-A)
Relation of asymmetric rewards to cost-sensitive learning left implicit	[11], [12]	Derivation from known results that the 5:1 reward equals the 5:1 cost-sensitive Bayes threshold (Eq. 3), and of how error-triggered termination departs from it; RL and cost-weighted models on one probability scale
RL detectors not compared with matched cost-sensitive or minimal baselines	–	Seven models with the same search procedure, the same number of configurations (16; 12 for logistic regression) and identical decision rules; two exposure-time baselines; group-bootstrap and exact sign-flip inference with multiplicity adjustment
Robustness assessed with optimization-based input attacks only	[4], [31]	Bounded, physically motivated telemetry perturbations as a sanity check, with adaptive evasion and false-alarm worst cases

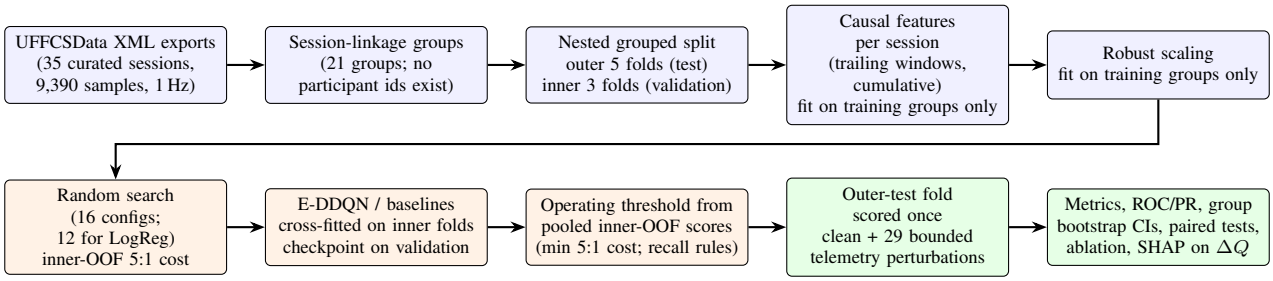


Fig. 1. Leakage-controlled, session-grouped evaluation pipeline. Blue: data construction (Steps 1–3; no step looks across a partition boundary); orange: model development using inner (validation) folds only (Steps 5–7); green: single evaluation of each outer-test fold with the frozen protocol (Steps 7–9).

TABLE II
MAIN TERMS AND NOTATION.

Term	Meaning
Session (episode)	One recorded play-through of one game, logged about once per second; one episode of the decision process.
Linkage group	Sessions that may come from the same person (Section III-B); the unit of every split.
Event second, y_t	$y_t = 1$ if the discomfort reported at second t is at least 1 (slight), otherwise $y_t = 0$.
State s_t	Causal feature vector at second t, computed from seconds $0, \ldots, t$ of the same session only.
Action a_t	1 = raise an alarm, 0 = stay silent.
Reward $R(a, y)$	Asymmetric reward of Eq. (1).
Regrets Δ_{FN}, Δ_{FP}	Reward lost by a missed event and by a false alarm (Eq. (4)).
$Q(s, a)$, $\Delta Q(s)$	Action value; decision margin $Q(s, 1) - Q(s, 0)$.
Score $\hat{p}(s)$	A model's output mapped to an estimate of $P(y = 1 \mid s)$ (Eqs. (5) and (7)).
5:1 cost c	$(5\,\mathrm{FN} + \mathrm{FP})/N$: expected cost per second when a miss costs as much as five false alarms.
OOF prediction	Out-of-fold prediction: the score of an inner-validation second from a model not trained on its group.
Operating rule	How the alarm threshold on $\hat{p}$ is chosen: D1, D0 or a recall constraint (Section III-G).

Nested grouped cross-validation. Every split assigns whole groups. The *outer* loop divides the groups into 5 folds; each fold serves once as test data, and the remaining groups form that fold's *development set*. The *inner* loop divides each development set into 3 folds: a model is trained on two inner folds and validated on the third, so that every development second receives exactly one out-of-fold (OOF) prediction. Inner validation data are used for hyper-parameter search, checkpoint selection and threshold selection; the outer-test fold is used only for the final evaluation. Folds are formed by stratified group k-fold splitting on game $\times$ label (outer seed 20261012; inner seeds 20261013–20261017). Every sample of the evaluation set therefore belongs to exactly one outer-test fold and is scored only by models that never saw its session, its linkage group or its exact questionnaire profile. Executable checks confirm zero session, group and profile overlap and zero duplicated feature rows between every development set and its test fold.

C. Step 3: Causal Features and Train-Only Preprocessing

A feature is *causal* if its value at second t uses only seconds $0, \ldots, t$ of the same session, as a monitor running during play would; we verified this by recomputing all features on sessions truncated at random times. Writing q_t for the three rotation components, the 39 telemetry features are:

- *instantaneous* (13): game indicator, elapsed time, speed, acceleration, |acceleration|, the three components of q_t, the per-second rotation change $\|q_t - q_{t-1}\|$, the frame rate and three region-of-interest indicators;
- *trailing windows* (22): over the last 5 s and the last 15 s (including t; shorter at the start of a session), the mean, standard deviation and change of speed, the mean and maximum of |acceleration|, the acceleration-spike rate,

the mean and standard deviation of the rotation change, and the mean, minimum and frame-drop fraction of the frame rate;
- *cumulative* (4): cumulative rotation change, cumulative numbers of spikes and frame drops, and the running mean speed.

A second is an *acceleration spike* if |acceleration| exceeds the 90th percentile of the training rows, and a *frame drop* if the frame rate is below 80 fps (more than 11 % below the 90-Hz display target). The logger writes a frame rate of 0 on the very first sample of some sessions; this artifact is replaced by the training median. In addition, 11 CSPQ indicators encode the seven questionnaire items (one indicator per non-reference answer, and one indicator per condition for vision problems); we call them the *questionnaire indicators*, and the feature-set names below refer to them as *profile*.

All learned preprocessing—the spike threshold, the imputation value and a median/IQR (robust) scaler—is fitted on the training rows of each inner split only and then applied unchanged to its validation and test rows. Columns that carry no usable information were removed: the field of view and the static-frame and auto-camera flags are constant within each game (pure game indicators), four configuration flags and posture are constant overall, and the absolute player position was excluded because it encodes elapsed course distance and is absent in the October wave. No oversampling or synthetic data are used anywhere; class imbalance is handled only inside training (Sections III-E and III-F).

Feature sets. Which features a model sees is a hyperparameter chosen on validation folds, from four sets: *telemetry* (39 features), *telemetry+profile* (50), *exposure* (8 exposure descriptors fixed a priori: game, elapsed time, cumulative rotation change, cumulative spikes, running and 15-s mean speed, and 15-s means of the rotation change and of |acceleration|) and *exposure+profile* (19). Two further sets, *time* (elapsed time only) and *time+game*, are used only by the post-hoc baselines of Section III-F.

D. Step 4: Decision Formulation—MDP or Contextual Bandit?

Decision process. As in DQN-based detection [11], [12], each session is an episode. At second t the detector observes the state $s_t \in \mathbb{R}^d$ (the causal feature vector), chooses an action $a_t \in \{0, 1\}$ (stay silent or raise an alarm) and receives the reward $r_t = R(a_t, y_t)$ with

$$\begin{aligned} R(1,1) &= +0.5, \quad R(0,0) = +0.1, \\ R(0,1) &= -0.5, \quad R(1,0) = -0.1. \end{aligned} \tag{1}$$

The next state s_{t+1} is the next second *of the same session*. The last second of a session is terminal ($d_t = 1$; otherwise $d_t = 0$), so no transition links two sessions.

Key observation. The telemetry was logged while participants played, with no alarm in the loop, so the action cannot change what is observed next: $P(s_{t+1} \mid s_t, a_t) = P(s_{t+1} \mid s_t)$. For any discount factor $\gamma < 1$, the optimal action-value function therefore splits into an immediate term and a term that does not depend on the action:

$$\begin{aligned} Q^*(s,a) = &\underbrace{\mathbb{E}[R(a, y_t) \mid s_t = s]}_{\text{immediate reward}} \\ &+ \underbrace{\gamma\, \mathbb{E}[(1 - d_t)\, V^*(s_{t+1}) \mid s_t = s]}_{\text{same for both actions}}. \end{aligned} \tag{2}$$

Three consequences follow.

(i) The problem is a contextual bandit. Because the second term of Eq. (2) is the same for both actions, the best action at every second is the one with the higher expected immediate reward, whatever the value of γ. The logged-data problem is therefore a *contextual bandit* [35]—a series of one-step decisions, each based on an observed context, whose actions do not affect later contexts—rather than a control problem in which actions steer future states.

(ii) The reward fixes a cost-sensitive threshold. With $p(s) = P(y = 1 \mid s)$, the decision margin of the optimal Q-function is

$$Q^*(s,1) - Q^*(s,0) = p(s)\, \Delta_{\mathrm{FN}} - (1 - p(s))\, \Delta_{\mathrm{FP}}, \tag{3}$$

where the *regrets*

$$\begin{aligned} \Delta_{\mathrm{FN}} &= R(1,1) - R(0,1) = 1.0, \\ \Delta_{\mathrm{FP}} &= R(0,0) - R(1,0) = 0.2 \end{aligned} \tag{4}$$

are the reward lost by missing an event and by raising a false alarm. The greedy policy alarms if and only if the margin is non-negative, i.e. if $p(s) \geq \Delta_{\mathrm{FP}}/(\Delta_{\mathrm{FN}} + \Delta_{\mathrm{FP}}) = 1/6$. This is exactly the Bayes-optimal decision of a classifier with misclassification costs $C_{\mathrm{FN}} : C_{\mathrm{FP}} = 5 : 1$ [17], which alarms when the expected cost of staying silent, $5p$, is at least the expected cost of an alarm, $1 - p$. The asymmetric reward is therefore not a different objective from cost-sensitive learning: it encodes the same 5:1 operating point.

(iii) All models can share one score scale. Solving Eq. (3) for p turns a learned Q-function into an implied probability,

$$\hat{p}(s) = \frac{\Delta Q(s) + \Delta_{\mathrm{FP}}}{\Delta_{\mathrm{FN}} + \Delta_{\mathrm{FP}}}, \qquad \Delta Q(s) = Q(s,1) - Q(s,0), \tag{5}$$

computed with each agent's own reward table (for the symmetric ± 1 reward of the standard DQN, $\Delta_{\mathrm{FN}} = \Delta_{\mathrm{FP}} = 2$). For a learned, imperfect Q-function, $\hat{p}$ is only an estimate: it is not calibrated by construction and can fall outside $[0, 1]$. Cost-weighted classifiers are mapped to the same scale by Eq. (7) (Section III-F), so the nominal Bayes cut-off is $1/6$ for every model. Two implementation choices move a model's score away from this scale. First, class-balanced replay (Section III-E) and class-balanced minibatches train on a class mix with a fraction $1-\rho$ (or one half) of events instead of the prevalence π of the training rows (0.44–0.46); the learned scores then estimate the posterior under that mix, whose odds are k times those of $P(y = 1 \mid s)$, with $k = \frac{1-\rho}{\rho} \cdot \frac{1-\pi}{\pi}$. The cut-off $\hat{p} \geq 1/6$ therefore corresponds to $P(y = 1 \mid s) \geq 1/(1 + 5k)$: for the selected configurations, 0.084–0.157 for E-DDQN, 0.120–0.208 for E-CB and 0.121–0.149 for the CS-MLP with balanced minibatches (Supplementary Sec. S5.12). Second, the Huber loss (Section III-E) estimates a conditional mean only

while residuals stay below its threshold of 1. This holds for the asymmetric reward at $\gamma = 0$, whose rewards span 0.6, but not for the ± 1 reward of the standard DQN, whose $\hat{p}$ is then a monotone but nonlinear function of $P(y = 1 \mid s)$: at $\gamma = 0$, $\hat{p} = 1/6$ corresponds to $P(y = 1 \mid s) = 0.25$; for $\gamma > 0$, large bootstrapped residuals have the same effect. Both shifts affect only the untuned cut-off (rule D0) and the greedy policy; the primary rule D1 chooses each model's threshold on its own validation scores, and ranking metrics do not depend on the scale. Consequences (i)–(iii) follow from standard results—the cost-sensitive Bayes threshold [17] and the equivalence of a contextual bandit with full reward information to cost-sensitive classification [19]; we state them because asymmetric-reward DQN detectors are usually presented without this link.

Error-triggered termination. Earlier DQN classifiers end an episode when a minority-class sample is misclassified [11], [12]. There the action does change the future: a missed event ends the reward stream, so the problem is no longer a contextual bandit. With $\gamma > 0$ a miss then also forfeits the discounted value of continuing, $\gamma V(s_{t+1})$, which acts as an additional, γ-dependent penalty on that error; the implied threshold therefore depends on γ and on the value of the remaining episode, not on the reward table alone. E-DDQN does not use error-triggered termination: its episodes end only with the recorded session, so Eqs. (2)–(5) apply to it, with $p(s)$ taken under the class mix of its replay (consequence (iii)).

Role of the discount factor. Bootstrapping with $\gamma > 0$ adds the action-independent second term of Eq. (2), whose estimation noise enters both Q-values; it cannot improve the optimal policy and can help or hurt only through estimation effects. We therefore treat $\gamma \in \{0, 0.5, 0.9, 0.99\}$ as a hyperparameter chosen on validation data and also evaluate the special case $\gamma = 0$ as a separately tuned model (E-CB). A genuine MDP would require the alarm to change later rendering or symptoms (e.g., a mitigation in the loop); no such intervention exists in the data, so we make no claim that E-DDQN learns adaptive sequential control.

E. Step 5: E-DDQN

E-DDQN combines the asymmetric reward of Eq. (1) with standard deep-RL components. Each component is listed with its purpose and its implementation.

- *Network.* A multilayer trunk (width and depth tuned; every hidden layer is linear, followed by batch or layer normalization [36], ReLU and, except after the last hidden layer, dropout [37]) feeds a dueling head [14], which separates a state value $V(s)$ from action advantages $A(s, a)$: $Q(s, a) = V(s) + A(s, a) - \frac{1}{2}\sum_{a'} A(s, a')$. The Q-values are not squashed.
- *Double-DQN target.* To reduce the over-estimation caused by the max operator [13], the online network θ chooses the next action and a target network θ^- evaluates it:

$$y_t = r_t + \gamma(1 - d_t)\, Q_{\theta^-}\big(s_{t+1}, \arg\max_{a'} Q_\theta(s_{t+1}, a')\big), \tag{6}$$

  where the arg-max is computed with the online network in evaluation mode; for $\gamma = 0$, $y_t = r_t$. The Huber loss [38], [39] (threshold 1) between $Q_\theta(s_t, a_t)$ and y_t is minimized with Adam [40] and ℓ_2 weight decay, gradient-norm clipping at 1.0 and a cosine learning-rate schedule [41] that decays to 10^{-5} over the training epochs.
- *Soft target updates.* After every gradient update, $\theta^- \leftarrow (1 - \tau)\theta^- + \tau\theta$ (Polyak averaging [42]); the batch-normalization running statistics are copied to the target network as well.
- *Class-balanced replay.* Transitions are stored in a circular buffer [43] of 50,000 transitions (doubled when full-information replay is used); every minibatch contains a fixed fraction ρ of non-event and $1 - \rho$ of event transitions, so that neither class dominates the updates; this also changes the class mix whose posterior the Q-values estimate (Section III-D).
- *Exploration and warm-up.* Actions are ε-greedy; ε starts at 1.0 and is multiplied by a tuned decay factor after every epoch, down to 0.01. Optionally, the buffer is first filled with 4,000 transitions with uniformly random actions.
- *Full-information replay (tuned on/off).* Offline, the label reveals the reward of *both* actions. E-DDQN may therefore also store the action that was not taken, together with *its own* reward. This is a valid transition, and it turns Q-learning into cost-sensitive regression of $\mathbb{E}[R \mid s, a]$.
- *Validation checkpointing.* Every second epoch the agent scores its inner-validation groups, and the epoch with the lowest achievable 5:1 cost (the minimum over thresholds; ties: higher PR-AUC) is kept.

Algorithm 1 summarizes training.

Implementation pitfalls. Three errors are easy to make when a classifier is trained as a Q-learner, and all three were present in an earlier implementation of this study: (i) building transitions from consecutive rows of shuffled data without terminal flags, so that the next state is an unrelated second, usually of another session; (ii) storing the true label as the action while computing the reward of a different, exploratory action (a form of teacher forcing that trains the correct action toward the penalty of the wrong one); and (iii) averaging only the parameters into the target network, so that its batch-normalization statistics never leave their initial values. Executable unit tests guard against each of them and also verify the 5:1 regret ratio and, on a toy problem, that the Q-values stay within 1.5 times the return bound.

F. Step 6: Matched Baselines

Every baseline receives the same splits, candidate feature sets, target, random-search procedure (16 configurations; 12 for logistic regression, whose search space contains only 24 distinct configurations), checkpoint rule (neural models), cross-fitted training recipe and threshold rules as E-DDQN. The number of configurations is matched, but not the coverage of the search spaces, which differ greatly in size (Section IV).

- **Standard DQN** [32]: symmetric ± 1 reward, plain (non-dueling) head, single max-operator target $r_t + \gamma(1 - d_t)\max_{a'} Q_{\theta^-}(s_{t+1}, a')$, hard target copies every 10,

Algorithm 1 E-DDQN training for one inner split

Initialize θ, $\theta^- \leftarrow \theta$, buffer $\mathcal{D}$; optional warm-up with random actions on within-session transitions
for epoch $= 1, \ldots, E$ **do**
  Concatenate training sessions in random order, each in temporal order; mark the last second of each session terminal
  for each chunk of 256 consecutive transitions **do**
    $a_t \leftarrow \varepsilon$-greedy on $Q_\theta(s_t, \cdot)$; $r_t \leftarrow R(a_t, y_t)$
    store $(s_t, a_t, r_t, s_{t+1}, d_t, y_t)$; if full-information replay, also store $(s_t, 1 - a_t, R(1 - a_t, y_t), s_{t+1}, d_t, y_t)$
    every 2 chunks: $k \in \{1, 2, 4\}$ (tuned) updates on class-balanced minibatches with target (6) and Huber loss; after each update, Polyak update of θ^- and copy of BN statistics
  end for
  5 consolidation updates; decay ε; cosine LR step
  every 2 epochs: score inner-validation groups; keep θ if the achievable 5:1 cost improves
end for
return checkpoint; score $\hat{p}(s)$ by Eq. (5)

100 or 500 updates (tuned), uniform replay and no full-information replay; it uses the same episodes, transitions, exploration and warm-up options, and γ is tuned as for E-DDQN.

- **E-CB**: E-DDQN with $\gamma = 0$, i.e. a contextual-bandit learner; the target network and the double estimator are then inactive by construction.
- **CS-MLP**: a cost-sensitive multilayer perceptron with the same trunk family and a positive-class-weighted cross-entropy ($w = 5$, i.e. the same 5:1 cost), trained on uniform or class-balanced minibatches and checkpointed like the agents.
- **Logistic regression**, **histogram gradient boosting (HGB)** and **random forest (RF)** [44], each with class weight 5 for event seconds.
- Two trivial references: *always alarm* and *never alarm*.

One score scale for weighted models. With event weight $w = 5$, a weighted classifier estimates $p_w = 5p/(5p+1-p)$ instead of $p = P(y = 1 \mid s)$. Inverting this relation gives

$$\hat{p} = \frac{p_w}{5 - 4p_w}, \tag{7}$$

so that, together with Eq. (5), every model's score is on the nominal scale of $P(y = 1 \mid s)$, with Bayes cut-off $1/6$. For the CS-MLP with class-balanced minibatches, the agents with class-balanced replay and the standard DQN, the scale is shifted as described in Section III-D.

Post-hoc exposure-time baselines. Because exposure duration is a well-documented driver of cybersickness [45], the labels are close to step functions within a session, and the primary analysis selected exposure features for most models, two minimal baselines were specified after the primary analysis and run under exactly the same splits, selection, threshold rules and seeds: **Time-LR**, a class-weighted logistic regression on elapsed time only (all six values of C), and **Time+game HGB**, gradient boosting on elapsed time and the game indicator (same search space and budget as HGB). Both are reported separately from the primary comparison.

G. Step 7: Evaluation Protocol and Statistics

The protocol was written and committed before any model was trained on the evaluation sessions; the post-hoc analyses (i)–(iv) listed at the end of this subsection were specified and committed after the primary results were known but before they were run (except the Holm adjustment in (ii)), the comparisons of the exposure-time baselines with the primary models, the Holm adjustment in (ii) and the Q-value diagnostic of Section V-G were added without prior commitment, analysis (v) was added after a pre-submission review, when all other results were known, analysis (vi), prompted by the same review, was specified and committed before any of its models was trained, and the descriptive analysis (vii) was added in the final pre-submission check. For each model family and outer fold, the protocol has four stages, and the outer-test fold enters only in the last one.

1) *Hyper-parameter search.* Each of the family's configurations (drawn once with a fixed seed) is trained on the three inner splits with one seed. The three validation predictions are pooled into OOF predictions for the whole development set, and the configuration with the lowest 5:1 cost $c = (5\,\mathrm{FN} + \mathrm{FP})/N$ at its cost-minimizing OOF threshold is selected (ties: higher PR-AUC).
2) *Final training.* The selected configuration is trained again with five seeds. For each seed, the three inner models are kept, each checkpointed on its own validation fold (*cross-fitting*): their validation predictions form the OOF scores, and the mean of their three scores is the outer-test score. The OOF and test scores therefore come from the same models.
3) *Operating thresholds*, chosen on the OOF scores only. The primary rule *D1* takes the threshold that minimizes the 5:1 cost (ties: the one closest to $1/6$). The secondary rules are *D0*, the untuned cut-off $\hat{p} \geq 1/6$ (the nominal 5:1 Bayes cut-off and, for the asymmetric agents, the greedy arg-max policy; its effective position on $P(y = 1 \mid s)$ for each model is given in Section III-D), and the largest threshold that reaches a validation recall of 0.95. Matched operating points use validation-chosen thresholds for recall targets of 0.80, 0.90, 0.95 and 0.99 and specificity targets of 0.5 and 0.8.
4) *Single test evaluation.* The outer-test fold is scored once with the frozen models and thresholds, on clean telemetry and on the perturbed telemetry of Section III-H.

Metrics. We report accuracy, precision, recall, specificity, F1, balanced accuracy, MCC [46], G-mean, false-positive and false-negative rates, missed events, the 5:1 cost, ROC-AUC and PR-AUC [47], exact confusion matrices, and mean $\pm$ SD over the five seeds. Event-level metrics complement the per-second metrics: the onset of a session is its first event second; an event counts as detected if an alarm is raised at or after onset; the detection delay is the time from onset to that alarm; and alarm fractions, the fraction of seconds with an alarm in a session averaged over sessions, are computed before onset and in sessions without any event.

Statistical inference. Analyses that need one prediction per

sample use the seed ensemble (Section IV-C). Ninety-five percent confidence intervals are percentile intervals from a bootstrap that resamples the 21 linkage groups (2,000 replicates). Paired comparisons use the same test samples. The primary paired test is a group bootstrap of the cost difference; its two-sided p-value is $2\min\{\Pr(\Delta^* \le 0), \Pr(\Delta^* \ge 0)\}$ over the replicates Δ^*, so with 2,000 replicates it is resolved only to about 0.001. The 9,390 test seconds come from only 35 sessions, 21 groups and 26 sessions with an onset, which is the relevant sample size for inference. With so few groups of unequal size, cluster-bootstrap inference is approximate and can be anti-conservative [48]; we therefore read p-values between 0.01 and 0.05 as approximate and also report an exact sign-flip test (analysis (v)). The intervals describe the sampling of test groups given the trained models; they do not include retraining variability beyond the five seeds. McNemar's test [49], [50] (exact binomial and continuity-corrected χ^2) is also reported, and its contingency table is checked against the error counts ($b-c = \mathrm{err}_B - \mathrm{err}_A$); because consecutive seconds of a session are strongly dependent, this sample-level test overstates the evidence and is secondary.

Post-hoc analyses. (i) Holm's step-down adjustment [51] of the six group-bootstrap p-values comparing E-DDQN with the other primary models, per operating rule. (ii) A paired group bootstrap of each model's cost against the always-alarm policy; its Holm adjustment over the nine models was added without prior commitment. (iii) Two grouping sensitivity analyses: a bootstrap that resamples the 35 sessions instead of the 21 groups, and a leave-one-group-out jackknife of every cost and cost difference. (iv) Validation-chosen cost curves: for each false-negative to false-positive cost ratio $R \in \{1, 1.5, 2, 3, 4, 5, 7, 10, 15, 20\}$, the threshold minimizing $R\,\mathrm{FN} + \mathrm{FP}$ on the pooled inner-OOF scores of each outer fold is applied to that fold's test data, and the test cost is divided by the cost of the better trivial policy at the same R (values below 1 mean that the model beats both always and never alarming). (v) From the saved predictions and search logs only: ROC-AUC computed within each outer fold and averaged over folds, next to the pooled ROC-AUC, which combines the scores of five fold-specific models; an exact two-sided sign-flip permutation test of each paired cost difference with the 21 groups as units (all 2^{21} sign patterns), Holm-adjusted like (i) and (ii); an exploratory comparison of the within-fold ROC-AUC of selected configurations with and without the questionnaire indicators; and a tally of the feature sets offered by each family's random-search draw, from the search logs. (vi) A search-draw sensitivity analysis of E-DDQN and the standard DQN: each is tuned again under the full protocol (same splits, budget, selection, checkpoint, cross-fitting, seeds and thresholds) with two further random draws of 16 configurations and with one draw restricted to the exposure feature set, and three decision rules fixed in advance state how the outcome bears on the comparison of the two agents, on the dependence of E-DDQN's result on its draw, and on the feature-set explanation of Section V-G (Supplementary Sec. S5.11). (vii) A descriptive calculation, without new models, of where the cut-off of rule D0 lies on $P(y = 1 \mid s)$ for each selected configuration, given its training class mix and loss (Section III-D; Supplementary Sec. S5.12).

H. Step 8: Telemetry-Perturbation Stress Tests

Threat model and scope. The motivating adversary controls the rendering or locomotion pipeline (e.g., through a malicious modification or a compromised runtime) and can degrade frame pacing (Type I) [3], distort camera or locomotion motion (Type II) or manipulate the field of view (Type III) to induce or exploit cybersickness [2], [4]. The dataset contains *no attacked sessions* and no measurement of what a manipulation would do to a participant. We therefore use perturbations only as a sanity check of which inputs the detectors rely on: synthetic, bounded changes of the logged telemetry of the outer-test sessions, after which the causal features are recomputed with the frozen, training-fitted featurizer and scaler, as a deployed monitor would see them. Labels are never changed.

Variants (fixed before evaluation; values in Supplementary Sec. S4.1). Frame-rate capping (Type I), orientation jitter, speed up-scaling and acceleration-spike injection (Type II), each at three strengths and applied to the whole session or to one random 60-s window, give 24 variants. Speed is clipped to each game's observed range and orientation to its valid range $[-1, 1]$; the frame-rate caps are applied as caps, so the 60- and 45-fps caps lie below every frame rate observed in Flight sessions after the first second (70–94 fps) and are out of the data range for that game. Field-of-view manipulation (Type III) is not evaluated, because the field of view never varies within a game in the data. Five further whole-session "hiding" manipulations (speed down-scaling, orientation smoothing and frame-rate restoration) are available only to an adaptive adversary. For each model and test session, this adversary picks among the 17 whole-session variants (12 stress scenarios and 5 hiding manipulations) the one that minimizes alarms on event seconds (*evasion*; the labels are known to it) or the one that maximizes alarms on non-event seconds (*false-alarm induction*). Choosing among fixed candidates gives lower bounds on vulnerability that are weaker than gradient-based attacks [4], [31]; the tests are feature-space stress tests, not demonstrations of physical rendering attacks.

I. Step 9: Explainability

We explain the decision margin $\Delta Q(s) = Q(s,1) - Q(s,0)$ of E-DDQN rather than a single Q-value, because $Q(s,1)$ alone also contains the state value $V(s)$, which is shared by both actions and does not explain the decision. SHAP values [52] are computed for one model per outer fold, retrained with the configuration, seed and inner split of that fold's first final model, with 200 background samples drawn from its training rows and attributions for 400 of its test samples. DeepExplainer [53] was tried first with its additivity check enabled; in every fold it failed that check (it does not recognize the layer-normalization modules of the selected networks), so the expected-gradients estimator of the SHAP library (GradientExplainer) was used instead. The method used in each fold is recorded with the released results.

IV. Experimental Setup

A. Implementation

All models are implemented in PyTorch [54] and scikit-learn [44] and run on CPU with one thread per process. The data parser, grouping, causal features, agents, baselines, perturbation generator, protocol runner and statistics are released with the paper; one script regenerates every number from the raw XML files.

B. Search Spaces and Budgets

The search spaces (Supplementary Sec. S2.1) are shared wherever a hyper-parameter is common to several families. Neural models: feature set (four options), trunk (64–64, 128–128, 256–256 or 256–512–256), dropout (0–0.3), normalization (batch or layer), learning rate (3×10^{-4}–3×10^{-3}), weight decay (0–10^{-3}) and batch size (256 or 512). Agents additionally: ε decay per epoch (0.90–0.97), warm-up (0 or 4,000 random-action transitions) and updates per learning step (1, 2 or 4); E-DDQN: $\gamma \in \{0, 0.5, 0.9, 0.99\}$, $\tau \in \{0.005, 0.01, 0.05\}$, replay composition $\rho \in \{0.4, 0.5, 0.6\}$ and full-information replay (on/off); E-CB: ρ and full-information replay; DQN: γ and the hard-update period (10, 100 or 500 updates); CS-MLP: uniform or class-balanced minibatches. All agents are trained for 40 epochs (one epoch is one pass over every training session in temporal order), with chunks of 256 transitions, a learning step every two chunks, five consolidation updates per epoch, a 50,000-transition buffer and gradient clipping at 1.0; the CS-MLP is trained for 40 epochs. Validation-based checkpoints are taken every two epochs. These budgets were calibrated on the October-2019 development-only sessions, where validation cost plateaued after 20–30 epochs. Each family receives 16 random configurations (12 for logistic regression), drawn once with a fixed seed and evaluated on every inner split of every outer fold. The same number of draws covers very different fractions of the search spaces: E-DDQN's space contains 2,985,984 distinct configurations, the DQN's 497,664, E-CB's 248,832 and the CS-MLP's 4,608, against 1,728 for gradient boosting, 192 for the random forest and 24 for logistic regression. The budget is therefore matched in the number of configurations, not in coverage; that a large RL search space is hard to explore with a small budget is itself a practical property of these detectors. Gradient boosting and random forests search learning rate, number of iterations, leaf counts, minimum leaf sizes, ℓ_2 regularization, depth and feature sub-sampling; logistic regression searches the inverse regularization strength $C \in [10^{-4}, 10]$. The configuration selected for each outer fold is listed in Supplementary Sec. S2.2.

C. Seeds, Seed Means and the Seed Ensemble

Final models are trained with five seeds per outer fold; for every seed the outer-test score is the average of the 3 cross-fitted inner models. We report mean $\pm$ SD over seeds. Paired tests, bootstrap confidence intervals, the comparison with always alarming, event-level metrics, matched operating points, perturbation results and cost curves need a single prediction per sample; for these we use the *seed ensemble*, i.e. the mean score over the five seeds, thresholded at the value chosen on the mean inner-OOF score. This choice was fixed in the evaluation code before the evaluation was run. Table IV reports both. They can differ, because thresholding an averaged score is not the same as averaging the metrics of separately thresholded seeds: for E-DDQN the ensemble has a higher 5:1 cost than the average seed (0.726 vs. 0.663), whereas for the standard DQN it is lower (0.536 vs. 0.584). Cost differences in the paired tests are therefore differences between seed ensembles and can be larger than the differences of seed means (E-DDQN vs. DQN: +0.190 vs. +0.079). No seed was selected or discarded.

D. Split-Protocol Comparison

To quantify how the split protocol affects measured performance, we evaluated the same 21 raw columns of the curated release under three protocols with an untuned gradient-boosting classifier: (i) oversampling every discomfort grade to the size of grade 0 (5,243 samples) *before* a random 70/15/15 row-level split, the protocol of the earlier, unpublished version of this study (it reproduces that version's partition sizes, 14,680 / 3,146 / 3,146, exactly); (ii) a row-level split without oversampling; and (iii) the grouped split of Section III-B. The earlier E-DDQN implementation was also run, unmodified, on partitions (i) and (iii). Because that version's 41-feature input files were not available, this is a reconstruction of its protocol rather than an exact reproduction.

V. Results

A. Why Session-Grouped Evaluation Is Required

This analysis motivates the evaluation design; it quantifies a known pitfall [15], [20] on these data and shows that a reconstruction of the earlier version's protocol yields similarly inflated scores, rather than introducing a new phenomenon. Table III evaluates the same 21 raw columns under three protocols. When every discomfort grade is oversampled to 5,243 samples before a 70/15/15 row split, an untuned gradient-boosting classifier reaches 97.6 % accuracy, 99.8 % recall, MCC 0.935 and ROC-AUC 0.995; 2,170 of the 2,359 positive test rows (92 %) have an exact duplicate in the training split, and every session contributes to both splits. Without oversampling, a row-level split still gives ROC-AUC 0.992 (accuracy 95.3 %), because adjacent seconds of the same session are nearly identical and carry the same label. With a session-grouped split, the same classifier falls to ROC-AUC 0.523 (MCC 0.005). The earlier E-DDQN implementation behaves the same way: 99.2 % recall and MCC 0.870 on the oversampled row-level partition, but MCC -0.004 on a grouped partition. Near-perfect scores on these data therefore measure the recognition of sessions seen during training, not the detection of cybersickness in new sessions, and they are obtained without any reinforcement learning. All results below use the leakage-controlled, session-grouped protocol of Section III-G; session-linkage groups reduce, but cannot exclude, leakage across participants (Section VI-E).

TABLE III
EFFECT OF THE EVALUATION PROTOCOL ON THE SAME DATA (21 RAW COLUMNS OF THE CURATED RELEASE; TEST-SPLIT METRICS). "PROXY" MARKS THE RECONSTRUCTION OF THE EARLIER PROTOCOL, WHOSE 41-FEATURE INPUT FILES WERE NOT AVAILABLE.

Model	Evaluation protocol	Acc.	Recall	Spec.	MCC	ROC-AUC
HGB	Oversample to 5,243/grade, then 70/15/15 row split (earlier protocol)	97.6	99.8	91.0	0.935	0.995
HGB	Row-level random split, no oversampling	95.3	94.1	96.3	0.905	0.992
HGB	Grouped 5-fold split (session-linkage groups)	52.2	33.6	66.9	0.005	0.523
E-DDQN (earlier code)	Earlier script, earlier protocol (proxy)	95.2	99.2	83.1	0.870	–
E-DDQN (earlier code)	Earlier script, grouped split (proxy)	41.1	92.1	7.7	−0.004	–

B. Session-Grouped Comparison under the 5:1 Cost (RQ2)

Table IV reports every model tuned, checkpointed and thresholded on validation groups only and scored once on the outer-test groups (all 9,390 samples, 35 sessions). Because the labels are close to step functions and these seconds come from only 21 independent groups and 26 sessions with an onset, the per-second metrics mainly measure how well a model reproduces when, and in which sessions, discomfort begins. Two facts dominate. First, the detection problem is hard once sessions and linkage groups are not shared: among the primary models, the best ROC-AUC is 0.692 (HGB) and 0.688 (random forest), where 0.5 corresponds to chance, and the best PR-AUC is 0.595 (standard DQN), where chance equals the prevalence of 0.44. Second, under a 5:1 cost the cost-minimizing operating point of every model lies close to "always alarm" (cost 0.558): validation-selected thresholds yield high recall but false-positive rates of 82–97 %. The random forest has the lowest cost, both as a mean over seeds (0.491±0.003) and as a seed ensemble (0.485); these are point estimates, and the differences among the supervised models were not tested.

With its primary search draw, E-DDQN has the highest cost and the lowest ROC-AUC of the 7 primary models (point estimates; tests in Section V-E; other draws in Section V-G): recall 93.4±2.5 %, specificity 7.4±2.7 %, cost 0.663±0.048 (above the 0.558 of always alarming as a point estimate; not significant, Section V-E), ROC-AUC 0.538±0.016 (seed ensemble 0.541, 95 % CI 0.382 to 0.690, which includes chance) and MCC 0.016; its seed ensemble is worse still (cost 0.726, recall 90.4 %), and its seed-ensemble cost is higher than the random forest's in 5 of 5 outer folds, by 0.004 to 0.727, i.e. only marginally in one fold (Supplementary Sec. S3.4). The standard DQN reaches cost 0.584 and ROC-AUC 0.661, and the contextual-bandit variant E-CB cost 0.575 and ROC-AUC 0.653. E-DDQN therefore does not improve on the standard DQN in this controlled comparison; with two further search draws it did not reach a significantly lower cost either, although it ranked seconds better than the DQN in one of them (Section V-G).

C. Exposure-Time Baselines

The post-hoc baselines (Table IV, Fig. 2) show what the primary models have learned. Time-LR, a logistic regression on elapsed time alone, reaches ROC-AUC 0.704 and PR-AUC 0.605, the highest pooled seed-mean values in Table IV. Computed within each outer fold and averaged, which avoids pooling the scores of five fold-specific models, the random forest ranks slightly higher (0.715 vs. 0.709), followed by HGB (0.702), and E-DDQN remains last (0.539; Supplementary Sec. S5.7) with its primary search draw, though not with other draws (up to 0.714; Section V-G). Time-LR's pooled ROC-AUC does not differ significantly from that of the random forest (difference +0.016, 95 % CI −0.071 to +0.102; group bootstrap); the interval is wide, so equivalence is not established either. At the 5:1 operating point, the random forest has a lower cost than Time-LR (seed ensembles 0.485 vs. 0.548; difference +0.063, CI +0.010 to +0.132; group bootstrap $p = 0.021$, exact sign-flip $p = 0.044$), with fewer missed events and fewer false alarms in the seed ensemble (Table IV). With 21 groups the bootstrap p-value is approximate and the exact sign-flip p-value lies just below 0.05, so the evidence that the additional inputs help the forest at the operating point is suggestive rather than conclusive. Adding the game indicator (Time+game HGB) gives cost 0.517 and ROC-AUC 0.676, again not significantly different from the random forest (cost difference +0.032, $p = 0.274$, sign-flip $p = 0.302$; ROC-AUC difference −0.012, $p = 0.760$). Both minimal baselines rank seconds better than the primary E-DDQN, Time-LR clearly and Time+game HGB less conclusively (seed-ensemble ROC-AUC differences +0.163, CI +0.063 to +0.287, group bootstrap p <0.001, and +0.135, CI +0.015 to +0.270, $p = 0.030$, which is approximate with 21 groups). On these data, therefore, no model was shown to discriminate better than a population onset curve, in which cybersickness becomes more likely the longer a session lasts, as long documented for exposure duration [45]; the wide intervals leave open modest gains that a sample of 21 groups cannot detect.

D. Operating Points, Cost Ratios and the Asymmetric Reward

Table V contrasts three operating rules. With the untuned cut-off $\hat{p} \geq 1/6$ (rule D0)—for the asymmetric agents the greedy policy $\arg\max_a Q(s, a)$, which because of class-balanced replay corresponds to $P(y = 1 \mid s) \geq 0.084$–0.157 for E-DDQN rather than to the 5:1 Bayes cut-off of $1/6$ (Section III-D)—E-DDQN obtains recall 77.2 % at specificity 28.0 % (cost 0.905), whereas the random forest obtains recall 94.5 % at specificity 34.9 % (cost 0.486). When every model is held to a validation recall of at least 0.95, the random forest again has the lowest cost (0.483; E-DDQN 0.785). At validation-matched recall targets (Table VI; descriptive, no test), the thresholds transfer worst for E-DDQN: at both recall targets shown, and also at the targets 0.80 and 0.99 (released results), it attains lower test recall than every supervised

TABLE IV

SESSION-GROUPED COMPARISON (RULE D1: THRESHOLD MINIMIZING THE 5:1 COST ON POOLED INNER-VALIDATION PREDICTIONS; 9,390 OUTER-TEST SAMPLES; RECALL, SPECIFICITY AND BALANCED ACCURACY IN %). LEFT: MEAN ± SD OVER FIVE SEEDS. RIGHT: THE SEED ENSEMBLE (MEAN SCORE OVER THE FIVE SEEDS, THRESHOLDED ON THE MEAN INNER-OOF SCORE; SECTION IV-C), WHOSE PREDICTIONS ARE USED FOR THE PAIRED TESTS, CONFIDENCE INTERVALS, COMPARISON WITH ALWAYS ALARMING AND COST CURVES; FN AND FP ARE EXACT COUNTS. BOLD: THE NAME OF THE PROPOSED MODEL AND THE LOWEST COST AMONG THE PRIMARY MODELS. HGB, LOGISTIC REGRESSION AND THE POST-HOC BASELINES ARE DETERMINISTIC GIVEN THE DATA, HENCE SD = 0 AND IDENTICAL LEFT AND RIGHT VALUES. PRECISION, F1, ACCURACY AND G-MEAN: SUPPLEMENTARY SEC. S3.1.

	Mean ± SD over five seeds							Seed ensemble (paired tests)				
Model	Recall	Spec.	Bal. acc.	MCC	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC	Recall	Spec.	$\text{Cost}_{5:1}$	FN	FP
E-DDQN	93.4±2.5	7.4±2.7	50.4±1.2	0.016±0.054	0.663±0.048	0.538±0.016	0.483±0.019	90.4	8.1	0.726	400	4,820
E-CB	97.1±2.9	8.4±2.8	52.8±1.2	0.122±0.052	0.575±0.055	0.653±0.011	0.562±0.006	96.6	14.6	0.552	140	4,480
DQN	95.2±2.7	14.5±4.9	54.8±1.6	0.158±0.038	0.584±0.040	0.661±0.013	0.595±0.022	96.9	16.3	0.536	129	4,391
CS-MLP	99.4±0.7	2.8±1.9	51.1±0.8	0.072±0.049	0.557±0.012	0.644±0.018	0.521±0.022	99.4	6.1	0.537	23	4,925
LogReg	99.1±0.0	8.1±0.0	53.6±0.0	0.167±0.000	0.532±0.000	0.626±0.000	0.576±0.000	99.1	8.1	0.532	36	4,817
HGB	97.0±0.0	14.6±0.0	55.8±0.0	0.197±0.000	0.543±0.000	0.692±0.000	0.593±0.000	97.0	14.6	0.543	124	4,479
RF	98.5±0.3	18.2±0.7	58.3±0.3	0.266±0.004	**0.491±0.003**	0.688±0.002	0.548±0.001	98.7	18.3	**0.485**	54	4,281
Post-hoc exposure-time baselines (added after the primary analysis)												
Time-LR (elapsed time)	96.7±0.0	15.0±0.0	55.8±0.0	0.194±0.000	0.548±0.000	0.704±0.000	0.605±0.000	96.7	15.0	0.548	138	4,457
Time+game HGB	97.4±0.0	17.6±0.0	57.5±0.0	0.239±0.000	0.517±0.000	0.676±0.000	0.550±0.000	97.4	17.6	0.517	107	4,319
Always alarm	100.0	0.0	50.0	0.000	0.558	–	–	100.0	0.0	0.558	0	5,243
Never alarm	0.0	100.0	50.0	0.000	2.208	–	–	0.0	100.0	2.208	4,147	0

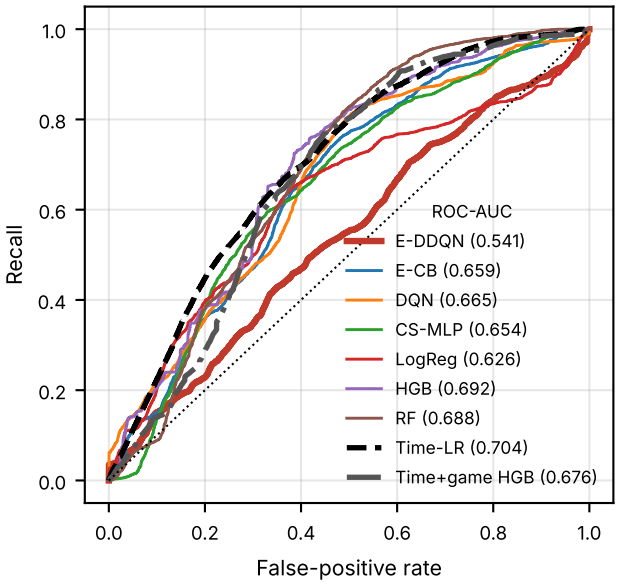


Fig. 2. ROC curves of the seed-ensemble scores, pooled over the five outer-test folds, for the seven primary models and the two post-hoc exposure-time baselines (dashed and dash-dotted). The legend gives the ROC-AUC of the seed-ensemble score, which can differ slightly from the seed means in Table IV.

model, and at the 0.95 target it also has a higher false-positive rate than all of them. Reading the ROC curves at fixed recall (descriptive only), the false-positive rate at 95 % recall is 97.7 % for E-DDQN and 65.8 % for the random forest. Because the reward encodes the same 5:1 operating point as a cost-weighted classifier (Eq. (3)), the asymmetric reward provided, on these data, no advantage that cost-weighted training plus a validation-selected threshold did not already provide.

The cost curves of Fig. 3 show how much the choice of the 5:1 ratio matters. At ratios of 1.5–2, where the cost-optimal policy is not close to a trivial one, the relative costs of all

TABLE V

EXPECTED 5:1 COST (MEAN OVER SEEDS) UNDER THREE OPERATING RULES CHOSEN WITHOUT TEST DATA.

Model	D1: min cost	D0: $\hat{p} \geq 1/6$	val. recall ≥ 0.95
E-DDQN	0.663	0.905	0.785
E-CB	0.575	0.663	0.581
DQN	0.584	0.680	0.573
CS-MLP	0.557	0.731	0.544
LogReg	0.532	0.808	0.518
HGB	0.543	0.567	0.511
RF	0.491	0.486	0.483
Always alarm	0.558	0.558	0.558

models except E-DDQN lie between 0.72 and 0.91 (random forest: 0.773 at $R = 2$). As R grows, the Bayes cut-off $1/(1+R)$ falls far below the prevalence, always alarming becomes nearly optimal, and the room for improving on it shrinks: at $R = 5$ the random forest's relative cost is 0.868 and at $R = 20$ it is 0.983. Models whose validation-chosen thresholds still miss event seconds on the test folds rise above 1 instead, because each miss is weighted by R: at $R = 20$ the relative cost is 1.508 for E-DDQN and 1.304 for the standard DQN. As point estimates, the random forest stays below the trivial policies at all 10 ratios and E-DDQN at 1 of them; at $R = 5$ neither model's difference from always alarming is significant after Holm adjustment (Section V-E). The negative safety result is therefore not an artifact of the particular ratio: over all models and ratios the lowest relative cost is 0.72 (random forest at $R = 1.5$), so no model removes more than about 28 % of the cost of the better trivial policy at any ratio.

E. Statistical Comparison

Table VII compares E-DDQN with every other primary model on identical test samples (seed ensembles, rule D1). Sample-level McNemar tests give $p < 0.001$ for every pair, but they treat 9,390 strongly autocorrelated seconds as independent and overstate the evidence; they are reported for completeness, with the contingency identity $b - c = \text{err}_B - \text{err}_A$

TABLE VI
MATCHED OPERATING POINTS: THRESHOLDS CHOSEN ON VALIDATION PREDICTIONS TO REACH THE STATED RECALL OR SPECIFICITY; TEST RECALL (%), FALSE-POSITIVE RATE (%) AND FALSE-POSITIVE COUNT OF THE SEED ENSEMBLE.

	val. recall ≥ 0.90			val. recall ≥ 0.95			val. specificity ≥ 0.80		
Model	Rec.	FPR	FP	Rec.	FPR	FP	Rec.	FPR	FP
E-DDQN	81.7	77.0	4,036	84.5	86.6	4,539	25.2	18.6	974
E-CB	88.6	68.0	3,565	92.4	78.1	4,093	31.2	17.4	910
DQN	88.0	64.2	3,366	91.7	72.7	3,810	32.2	14.2	743
CS-MLP	94.1	78.4	4,111	97.5	85.1	4,463	24.2	12.7	666
LogReg	92.8	73.5	3,856	97.2	81.7	4,285	37.7	22.5	1,181
HGB	95.4	72.2	3,785	97.5	81.7	4,282	24.3	11.3	594
RF	93.7	64.0	3,358	96.9	74.3	3,896	21.0	13.3	695

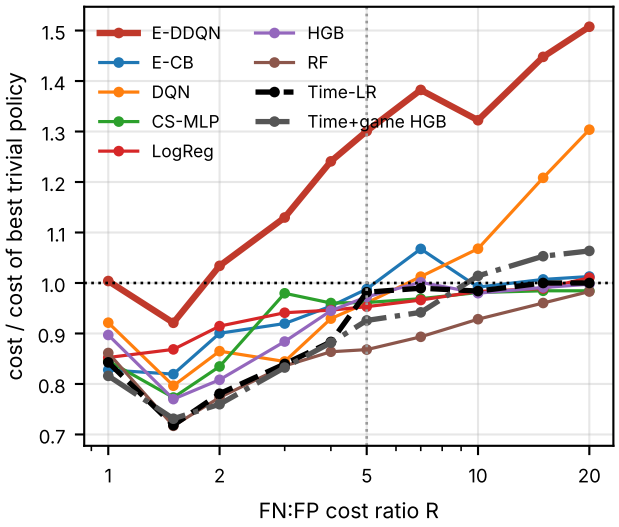


Fig. 3. Validation-chosen cost curves (post hoc): for each false-negative to false-positive cost ratio R, the threshold minimizing $R\,\mathrm{FN}+\mathrm{FP}$ on the inner-validation predictions is applied to the outer-test fold; the pooled test cost of the seed ensemble is divided by the cost of the better trivial policy (always or never alarm). Values below 1 improve on the trivial policy; the dotted vertical line marks the 5:1 ratio.

checked and recomputed with an independent library. The primary inference is the group bootstrap, which resamples the 21 independent units. E-DDQN's seed-ensemble cost (0.726) exceeds that of every other primary model by +0.175 to +0.242. These ensemble differences are larger than the corresponding differences of seed means in Table IV (+0.079 to +0.172) because E-DDQN's ensemble is worse than its average seed (Section IV-C); after Holm adjustment over the six comparisons, only the difference from the random forest remains significant under the group bootstrap (+0.242, 95 % CI +0.052 to +0.528; adjusted $p = 0.006$), whereas the adjusted p-values for the other five comparisons lie between 0.070 and 0.105. In particular, the difference from the standard DQN (+0.190; seed means +0.079) is not significant (unadjusted $p = 0.079$).

The direction of these conclusions is robust to the resampling unit and to individual groups, but the strength of the evidence depends on the test. Resampling the 35 sessions instead of the groups gives the same pattern (E-DDQN vs. random forest: CI +0.055 to +0.503, Holm-adjusted $p = 0.006$). In a leave-one-group-out jackknife over the 21 groups, the cost difference between E-DDQN and the random forest ranges from +0.137 to +0.284 and never changes sign; the same holds for every other comparison with E-DDQN. The exact group sign-flip test, which does not rely on a bootstrap approximation with 21 clusters [48], is more conservative: it gives $p = 0.014$ for E-DDQN versus the random forest (0.083 after Holm adjustment) and $p \geq 0.048$ for the other five comparisons (Supplementary Sec. S5.8). E-DDQN's higher cost than the random forest is therefore consistent across folds, groups and resampling schemes, but it is not established at a family-wise 5 % level by every test. Compared with always alarming, the random forest's seed-ensemble cost differs by −0.074 (CI −0.135 to −0.019; unadjusted $p = 0.012$), but after Holm adjustment over the nine models this is not significant ($p = 0.108$; sign-flip test $p = 0.022$, adjusted 0.194); E-DDQN's seed-ensemble cost exceeds that of always alarming by +0.168 (CI −0.024 to +0.466; unadjusted $p = 0.104$), which is not significant either: E-DDQN is not shown to be worse than the trivial policy, and, with linkage groups as the unit of inference, no model is shown to improve on it at the 5:1 ratio. Only when the 35 sessions rather than the groups are resampled, which ignores possible dependence between linked sessions, does the random forest's advantage remain significant after Holm adjustment ($p = 0.002$, adjusted 0.018, approximate). All comparisons in this subsection use E-DDQN's primary search draw; Section V-G reports how its comparison with the standard DQN changes with the draw.

F. Event-Level Behavior

At the cost-optimal operating points (seed ensembles) every model flags almost every session. Averaged over sessions, E-DDQN raises an alarm on 96.6 % of the seconds of each of the 9 sessions in which no discomfort was ever reported and on 82.3 % of the seconds before onset in the 26 sessions with an event (random forest: 75.3 % and 61.3 %). Events are therefore "detected" with zero median delay by construction, not through discrimination.

G. Why E-DDQN Under-Performs in the Primary Analysis

Validation selected $\gamma = 0.99, 0.99, 0.99, 0.99, 0.9$ for the five outer folds of E-DDQN, i.e. strong bootstrapping, and the exposure+profile feature set, which includes the questionnaire indicators, in 5 of 5 folds. Three exploratory diagnostics and

TABLE VII
PAIRED COMPARISONS WITH E-DDQN (A) ON THE SAME 9,390 TEST SAMPLES (SEED ENSEMBLES, RULE D1). b: A CORRECT, B WRONG; c: A WRONG, B CORRECT. ΔCOST IS THE DIFFERENCE OF THE SEED-ENSEMBLE COSTS (TABLE IV, RIGHT BLOCK); ITS CI AND p_{GROUP} COME FROM 2,000 GROUP-BOOTSTRAP REPLICATES (UNADJUSTED; HOLM-ADJUSTED VALUES IN THE TEXT); p_{EXACT} IS THE SAMPLE-LEVEL EXACT MCNEMAR TEST, WHICH IGNORES WITHIN-SESSION DEPENDENCE.

B (vs. A = E-DDQN)	b	c	err_A	err_B	χ^2_{cc}	p_{exact}	$\Delta\text{Cost}_{5:1}$ (A−B) [95% CI]	p_{group}
E-CB	299	899	5,220	4,620	299.5	<0.001	+0.175 [+0.016, +0.448]	0.014
DQN	297	997	5,220	4,520	377.6	<0.001	+0.190 [−0.017, +0.493]	0.079
CS-MLP	402	674	5,220	4,948	68.3	<0.001	+0.190 [+0.005, +0.485]	0.039
LogReg	389	756	5,220	4,853	117.0	<0.001	+0.194 [+0.011, +0.477]	0.026
HGB	203	820	5,220	4,603	370.9	<0.001	+0.183 [+0.011, +0.460]	0.035
RF	293	1,178	5,220	4,335	531.2	<0.001	+0.242 [+0.052, +0.528]	0.001

one experiment specified before it was run bear on why E-DDQN under-performs in the primary analysis.

(i) Bootstrapping. Section III-D shows that bootstrapping cannot improve the optimal policy here and only adds action-independent estimation noise to ΔQ. The separately tuned $\gamma = 0$ variant (E-CB) ranks better (ROC-AUC 0.653 vs. 0.538; its cost difference from E-DDQN is not significant after Holm adjustment, Section V-E), but it also selected different feature sets (no questionnaire indicators in 4 of 5 folds; Supplementary Sec. S2.2) and different networks, so this comparison does not isolate γ. With E-DDQN's configuration held fixed, setting $\gamma = 0$ raises ROC-AUC only from 0.538 to 0.576 and changes the cost by less than one seed standard deviation (Table IX).

(ii) Questionnaire indicators. Across the seven primary models, the 13 of 35 selected configurations (model–fold pairs) whose feature set includes the questionnaire indicators have a lower within-fold ROC-AUC than the 22 that do not (mean deviation from the fold mean −0.069 vs. +0.041; without E-DDQN, −0.056 vs. +0.020). In 4 of 5 outer folds every configuration with the indicators ranks below every configuration without them (Supplementary Sec. S5.9). A test group never shares its exact profile with the training groups, so these indicators can transfer only through shared attribute values; with 21 groups they may instead let a model fit group-level label rates that do not generalize. The analysis is exploratory and observational—feature sets were selected, not assigned—so it suggests, but does not establish, a cause.

(iii) Search draw and selection optimism. Each family's random configurations (16; 12 for logistic regression) were drawn once and reused in every outer fold. For E-DDQN this draw contained only 2 configurations with the exposure feature set without questionnaire indicators (E-CB 6, DQN 6) and 10 with the indicators (Supplementary Sec. S5.10). On validation data the best E-DDQN candidate with the indicators had the lower cost in 5 of 5 folds (e.g. 0.424 vs. 0.487 in fold 0), so the selection rule chose it each time. Experiment (iv) tests how much E-DDQN's result depends on which configurations its random draw happened to offer. Table VIII compares, per family, the inner-validation cost of the configuration at the moment it was selected with its outer-test cost. The gap is +0.288 for E-DDQN but between −0.015 and +0.070 for all other models. The selection criterion, the lowest achievable 5:1 cost on the inner out-of-fold predictions, is nearly flat near the always-alarm regime (0.42–0.54 for the selected configurations, against 0.558 for always alarming). Combined with noisy, high-variance training curves and validation checkpointing, it lets configurations look best on validation folds by chance [55].

(iv) Search-draw experiment. Specified, with three decision rules, and committed before any of its models was trained (Supplementary Sec. S5.11), this experiment tuned E-DDQN and the standard DQN again under the full protocol with two further random draws of 16 configurations each and with one draw restricted to the exposure feature set. In no draw did E-DDQN have a significantly lower cost than the standard DQN (seed-ensemble differences −0.135 to +0.190; smallest group-bootstrap $p = 0.079$), so the conclusion of Section V-B is not challenged (rule R1). E-DDQN's own result, however, depended strongly on the draw (rule R2): its seed-mean cost ranged from 0.549 to 0.663 and its within-fold ROC-AUC from 0.539, the lowest of the primary models, to 0.714, close to the highest (random forest, 0.715); the standard DQN varied less (cost 0.580–0.634, within-fold ROC-AUC 0.629–0.666). The ranking of the two agents reversed between draws: the standard DQN ranked seconds better in draw 1 (seed-ensemble ROC-AUC difference −0.125, CI −0.264 to −0.018, $p = 0.021$, approximate) and E-DDQN in draw 2 (+0.088, CI +0.021 to +0.162, $p = 0.009$). E-DDQN's further draws offered 8 and 7 exposure-only candidates, against 2 in draw 1, and selected the questionnaire indicators in 0 and 2 of 5 folds, against 5. Restricted to the exposure features, E-DDQN's within-fold ROC-AUC (0.687) was within 0.026 of the standard DQN's (0.712) and their cost difference was not significant (−0.025, CI −0.061 to +0.025), which supports the feature-set explanation (rule R3). In draw 2, E-DDQN selected $\gamma = 0.99$ in 3 of 5 folds and still reached its highest within-fold ROC-AUC, so strong bootstrapping did not by itself prevent good ranking (descriptive).

Together, the diagnostics and the experiment indicate that E-DDQN's weak primary result is largely a consequence of its search draw—a draw dominated by questionnaire-based configurations, combined with noisy validation-based selection—rather than of its reward or of bootstrapping. The experiment varied the draw and the feature set, not single components, so it does not show whether any component helps a configuration that ranks well. Q-value estimates stay finite, so the primary result is not caused by numerical divergence, but they are poorly scaled. The largest magnitude over all E-DDQN runs is 61.78 (DQN 10.10, E-CB 0.95), and relative to the return bound $r_{\max}/(1-\gamma)$ of each fold's selected discount the largest

TABLE VIII
SELECTION OPTIMISM: POOLED INNER-VALIDATION 5:1 COST OF THE SELECTED CONFIGURATION (HPO SEED, AT ITS COST-MINIMIZING THRESHOLD) VERSUS ITS OUTER-TEST COST (SEED ENSEMBLE, RULE D1). BOTH ARE AVERAGED OVER THE FIVE OUTER FOLDS; THE OUTER-TEST VALUES ARE THEREFORE UNWEIGHTED FOLD MEANS AND DIFFER FROM THE POOLED COSTS IN TABLE IV.

Model	Inner-OOF cost at selection	Outer-test cost	Gap
E-DDQN	0.467	0.754	+0.288
E-CB	0.488	0.557	+0.070
DQN	0.478	0.544	+0.065
CS-MLP	0.501	0.534	+0.033
LogReg	0.518	0.531	+0.014
HGB	0.518	0.539	+0.021
RF	0.495	0.481	−0.015

estimates exceed the bound by up to a factor of 2.3 for E-DDQN (in 2 of 5 folds), 2.2 for DQN and 1.9 for E-CB, i.e. extrapolation on atypical states. The mean greedy-action Q-value differs from its realized discounted validation return by −2.900 for E-DDQN, +0.393 for DQN and +0.088 for E-CB; for E-DDQN and E-CB, part of this difference reflects the class mix of the replay, which differs from that of the validation data (Section III-D).

H. Component Ablation and Reward-Ratio Sensitivity

Table IX removes or changes one component of E-DDQN at a time while keeping the configuration selected for each outer fold fixed (five seeds; same protocol). No formal test was performed, so the differences below are point estimates. Because the selected configurations already used no warm-up in four of the five folds, the "no warm-up" variant differs from the full model only in the remaining fold. This analysis and the reward-ratio analysis below hold E-DDQN's primary (draw-1) configurations fixed, which rank weakly (Section V-G); they cannot show whether a component helps a configuration that ranks well. Every variant ranks seconds only weakly (ROC-AUC 0.529–0.576, where 0.5 is chance). No single change moves the cost by more than about one seed standard deviation of the full model (0.663±0.048). The lowest cost is obtained by *replacing* the asymmetric reward with the symmetric ±1 reward (0.625) and the highest ROC-AUC by switching off bootstrapping ($\gamma = 0$; 0.576); hard target copies (0.634), no dueling head (0.644), uniform replay (0.660), no warm-up (0.656) and toggling full-information replay (0.667) leave the cost essentially unchanged, and removing the double estimator slightly increases it (0.681). None of these components can be credited with a gain in safety or accuracy. Every variant is scored on the same implied-probability scale and thresholded with the same rules, so the reward table can act only through the learned scores; at the fixed cut-off $\hat{p} \geq 1/6$ (rule D0) the symmetric-reward variant is not worse either (cost 0.825 vs. 0.905).

Table X varies the false-negative to false-positive regret of the reward from 1:1 to 10:1 (E-DDQN configuration per fold fixed; ratio 5:1 reproduces the main E-DDQN run bit for bit, which also confirms the determinism of the pipeline). The greedy policy moves along the trade-off in the direction Eq. (3) predicts (its nominal cut-off is $1/(1+R)$; because of class-balanced replay it corresponds to $1/(1+kR)$ on $P(y=1 \mid s)$, with $k = 1.31$–1.95 for the selected E-DDQN configurations, Section III-D): recall rises from 52.4 % (1:1) to 77.2 % (5:1) and 80.7 % (10:1), while specificity falls from 58.1 % to 28.0 % and 23.0 %. Once the threshold is chosen on validation data, however, all ratios end at similar costs (0.629–0.673; the largest difference is smaller than any ratio's seed standard deviation, and no formal test was performed) and similar ranking: the reward ratio sets the default operating point, but it does not create discrimination that a threshold cannot reproduce. The 5:1 ratio is therefore best understood as a declared safety preference whose operating point can equally be reached by cost-weighted training or threshold selection.

I. Sanity Check: Telemetry Perturbations (RQ3)

Table XI reports threshold-free ranking under the whole-session perturbations. 87 % of the model–scenario combinations change the pooled ROC-AUC by less than 0.03 and none by more than 0.089 (largest drops: acceleration-spike injection in E-CB and DQN, speed down-scaling in gradient boosting). This insensitivity is not evidence of a robust safety monitor. Of the 35 selected configurations, 29 use the exposure feature sets, in which elapsed time and the game indicator are untouched by every perturbation and the remaining inputs are averages or cumulative sums over many seconds; a perturbation applied to every test session alike can, moreover, shift the scores without reordering them; and the primary E-DDQN's ranking is not distinguishable from chance in any case (Section V-B). Fixed thresholds are more informative: at the cost-optimal operating points (Supplementary Sec. S4.3), the per-session worst-case evasion adversary lowers recall on event seconds by 4–19 percentage points (E-DDQN: from 90.4 % to 82.9 %), and the false-alarm adversary raises every model's false-positive rate to 94.5–100 % (81.7–93.9 % for the seed ensembles without manipulation). These are lower bounds from a weak, candidate-selecting adversary on feature-space perturbations, not evidence about real rendering attacks.

J. Explainability

SHAP attributions of the decision margin ΔQ of the primary E-DDQN (Fig. 4) are dominated by *cum_rot*, *elapsed_s*, *speed_mean_sofar*, *CSPQ VR experience: some*, *cum_acc_spikes* (feature names as in the released feature dictionary, Supplementary Sec. S1.4). The decisions are driven mainly by exposure time, cumulative exposure and questionnaire answers rather than by instantaneous telemetry, consistent with the exposure-time baselines and with diagnostic (ii) of Section V-G. Because the primary E-DDQN's ranking is not distinguishable from chance (Section V-B), these attributions describe what the model uses, not a validated signal. Because attributions are computed after the decision, they add no latency to detection.

VI. DISCUSSION

A. Answers to the Research Questions

Evaluation protocol. On these data the split protocol, not the model, decides whether scores look near-perfect: random

TABLE IX
LEAVE-ONE-COMPONENT-OUT ABLATION OF E-DDQN (SELECTED CONFIGURATION PER OUTER FOLD HELD FIXED; RULE D1; MEAN OVER 5 SEEDS, COST ± SD). THE NO-WARM-UP VARIANT CHANGES ONLY THE ONE FOLD WHOSE SELECTED CONFIGURATION USED WARM-UP.

Variant	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC	PR-AUC
Full E-DDQN (selected)	93.4	7.4	0.663±0.048	0.538	0.483
full-information replay toggled	93.7	5.4	0.667±0.106	0.563	0.500
symmetric ±1 reward instead of asymmetric	94.8	8.6	0.625±0.038	0.572	0.517
uniform instead of class-balanced replay	94.3	4.3	0.660±0.045	0.554	0.486
no double estimator (max-operator target)	93.1	5.3	0.681±0.086	0.543	0.491
no dueling head	94.6	5.8	0.644±0.059	0.559	0.494
hard target copy every 10 updates	94.9	6.5	0.634±0.043	0.529	0.472
no warm-up	93.8	7.1	0.656±0.049	0.547	0.497
$\gamma = 0$ (contextual bandit)	94.4	6.4	0.645±0.070	0.576	0.510

TABLE X
REWARD-RATIO SENSITIVITY OF E-DDQN: OPERATING POINT OF THE GREEDY (ARG-MAX) POLICY, WHOSE NOMINAL CUT-OFF IS $1/(1+R)$ ($1/(1+kR)$ ON $P(y=1 \mid s)$ UNDER CLASS-BALANCED REPLAY; SECTION III-D), AND OF THE VALIDATION-TUNED THRESHOLD (RULE D1). MEAN (± SD) OVER FIVE SEEDS.

		Greedy policy ($\arg\max_a Q$)			Validation-tuned threshold (D1)			
FN:FP regret	implied cut-off	Recall	Spec.	$\text{Cost}_{5:1}$	Recall	Spec.	$\text{Cost}_{5:1}$	ROC-AUC
1:1	$1/2$	52.4±5.6	58.1±4.8	1.285	94.9	7.4	0.629±0.056	0.558
2:1	$1/3$	69.2±3.3	43.1±1.6	0.999	94.6	6.4	0.642±0.074	0.563
5:1	$1/6$	77.2±3.8	28.0±2.0	0.905	93.4	7.4	0.663±0.048	0.538
10:1	$1/11$	80.7±3.2	23.0±3.3	0.855	92.9	7.4	0.673±0.095	0.552

TABLE XI
THRESHOLD-FREE ROBUSTNESS: POOLED OUTER-TEST ROC-AUC OF THE SEED-ENSEMBLE SCORE ON CLEAN TELEMETRY AND UNDER WHOLE-SESSION PERTURBATIONS (THE LAST COLUMN GIVES THE MOST NEGATIVE AND MOST POSITIVE CHANGE OVER ALL 17 WHOLE-SESSION VARIANTS, INCLUDING THE EVASION-ONLY CANDIDATES).

Model	Clean	fps≤45	jitter 0.10	speed ×1.5	spikes 40	speed ×0.5	smooth 0.25	min / max Δ
E-DDQN	0.541	0.541	0.593	0.561	0.559	0.532	0.539	−0.008 / +0.053
E-CB	0.659	0.659	0.682	0.648	0.575	0.648	0.658	−0.089 / +0.052
DQN	0.665	0.665	0.677	0.680	0.605	0.632	0.676	−0.064 / +0.035
CS-MLP	0.654	0.649	0.654	0.630	0.652	0.629	0.646	−0.025 / +0.005
LogReg	0.626	0.677	0.639	0.620	0.629	0.608	0.639	−0.017 / +0.051
HGB	0.692	0.694	0.709	0.650	0.698	0.647	0.712	−0.068 / +0.019
RF	0.688	0.689	0.708	0.667	0.683	0.723	0.725	−0.021 / +0.037

row-level splits—with or without oversampling—give ROC-AUC above 0.99 because adjacent seconds of a session are near-duplicates with the same label, whereas session-grouped evaluation of the same columns gives 0.523. This is a known pitfall rather than a new finding, but it means that accuracy figures on gameplay cybersickness data should be read together with the split protocol. *RQ1.* On logged data the asymmetric reward encodes a cost-sensitive operating point: the greedy policy is the Bayes rule $P(y=1 \mid s) \geq 1/6$ of 5:1 cost-sensitive classification, and bootstrapping cannot improve it (Section III-D). *RQ2.* Under leakage-controlled evaluation with the same search procedure and decision rules, E-DDQN does not have a significantly lower 5:1 cost than the standard DQN, with its primary search draw or with two further draws specified before they were run; in ranking, the standard DQN was better with the primary draw ($p = 0.021$, approximate) and E-DDQN with another ($p = 0.009$), so neither agent is consistently better. With its primary draw its seed-ensemble cost is also higher than that of a cost-weighted random forest in every outer fold (marginally in one; significant under the group bootstrap, but not after Holm adjustment under the more conservative exact sign-flip test) and ranks seconds worse than elapsed time alone; that draw was the weakest of its three, and with the others its within-fold ROC-AUC reached 0.714 (not compared with these models by a test). No primary model ranked seconds significantly better than elapsed time, and no model meets a safety-prioritized target. *RQ3.* As a sanity check, bounded manipulation of the telemetry channels changes 87 % of the scenario-wise ROC-AUC values by less than 0.03 (at most 0.089), which is not evidence of robustness (Section V-I), while an adaptive adversary with label knowledge can still lower recall at fixed thresholds by 4–19 percentage points.

B. Is E-DDQN Reinforcement Learning?

With logged data the alarm cannot influence the subsequent telemetry, so the decision problem is a contextual bandit. E-DDQN remains a valid value-based learner of a cost-sensitive detection policy, but (i) the asymmetric reward is mathematically equivalent to a 5:1 cost-sensitive Bayes threshold, so it is a statement about the operating point rather than a different learning objective, and the class mix of the replay

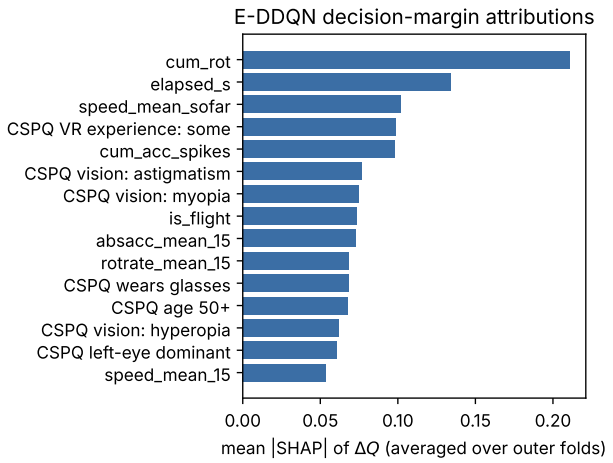


Fig. 4. Mean absolute SHAP value of the E-DDQN decision margin ΔQ (primary search draw), averaged over the five outer folds (one model per fold; attributions on that fold's test groups).

shifts that operating point as well (Section III-D); (ii) bootstrapping cannot improve the optimal policy, and empirically removing it with E-DDQN's configuration held fixed changed little, whereas E-DDQN's weak primary result was largely a consequence of its random search draw, which was dominated by configurations that use the questionnaire answers (Section V-G); and (iii) under bandit feedback the agent observes the reward of one action only, although labels reveal both, which is why full-information replay is the principled way to use the labels. The same reasoning applies to any DQN-based classifier trained on a fixed dataset whose episodes end only with the data: its benefit has to come from estimation effects, and it should be compared with cost-weighted supervised models at the same operating point. Formulations that end an episode when a minority-class sample is misclassified [11], [12] are action-dependent, but the termination acts only as an additional, γ-dependent penalty on that error (Section III-D), so the same comparison is needed to show that they add more than a different operating point. Adaptive sequential control would require an intervention in the loop—for example an alarm that triggers a rendering mitigation whose effect on later telemetry and symptoms is measured—which, to our knowledge, no public cybersickness dataset provides.

C. The Safety Objective

A 5:1 cost makes an alarm the cheaper decision whenever $P(y = 1 \mid s) \geq 1/6$. With a prevalence of 0.44 and weakly informative inputs, almost every second satisfies this, so the cost-optimal policy of every model is close to "always alarm": recall is high, but 82–97 % of non-event seconds also trigger an alarm. High recall alone is therefore not evidence of a useful safety monitor. Judged on cost, specificity at matched recall, threshold-free ranking and cost curves across ratios, the intended safety objective—near-complete detection with a controlled false-alarm rate—is not achieved by any model on these data. The exposure-time baselines suggest where the obstacle lies: once sessions and linkage groups are not shared between training and test, no primary model ranked seconds significantly better than how long the session has lasted, which distinguishes individual susceptibility only coarsely.

D. The Adversarial Threat Model

Rendering manipulations that could induce sickness or physical harm have been described [2], [3], and optimization-based input attacks can disable deep cybersickness detectors [4], [31]. UFFCSData, however, contains no manipulated sessions, so its label marks cybersickness, not an attack. Our stress tests therefore answer a narrower question than the cited attack studies: how far bounded, physically motivated changes of the logged frame-rate, orientation and locomotion channels move the detectors' decisions. The small effect on ranking is not evidence of a robust design: it reflects the inputs that the selected configurations use and the fact that a perturbation applied to every test session alike can shift scores without reordering them, while at fixed thresholds an adaptive adversary still changes decisions. A detector that used the telemetry more would also be easier to manipulate through it. Demonstrating detection of real rendering attacks requires data in which manipulations are applied to participants under ethical approval and their physiological effects are measured.

E. Limitations

(i) A single public dataset with 35 curated sessions, 21 independent groups and 26 onsets; confidence intervals are wide, fold-to-fold variation is large, and inference with so few clusters is approximate: the group bootstrap and the exact sign-flip test agree in direction but differ in strength (E-DDQN vs. random forest, Holm-adjusted $p = 0.006$ vs. 0.083). (ii) Participant identity cannot be verified; linkage groups are conservative but cannot exclude a participant who answered the questionnaire differently in non-consecutive sessions, and they may merge different people with identical profiles. (iii) The label is a coarse self-report, logged at 1 Hz but changing rarely within a session, whose elicitation procedure (prompted or spontaneous reports) is not documented, and the binary threshold (discomfort ≥ 1) merges slight with severe discomfort; a safety monitor might target grades 2–3 only, which are almost absent from Flight (41 of 5,397 samples) although common in Race (1,450 of 3,993). (iv) All analyses beyond the frozen protocol are post hoc. The exposure-time baselines, the Holm adjustment, the comparison with always alarming, the grouping sensitivity analyses and the cost curves were specified after the primary results were known and committed before being run; the paired comparisons of the exposure-time baselines with the primary models, the Holm adjustment of the comparison with always alarming and the Q-value-scale diagnostic were added without prior commitment; the within-fold ROC-AUC, the sign-flip tests and the feature-set and search-draw tallies were added after a pre-submission review, when all other results were known; and the search-draw experiment, prompted by that review, was specified and committed before it was run; and the descriptive score-scale calculation of rule D0 (Section III-D) was added in the final pre-submission check. (v) Perturbations are synthetic

and feature-space, their physiological effect is unknown, and the adversary is weaker than optimization-based attacks. (vi) Hyper-parameter search used a modest budget (16 configurations per family; 12 for logistic regression) that covers a much smaller fraction of the RL search spaces (about 3.0 million configurations for E-DDQN) than of the random-forest space (192); at this budget, E-DDQN's result depended strongly on the random draw and the standard DQN's less (Section V-G). The draws of the other five families were not varied, so their results, and the ranking among all models, may also depend on their draws, and a larger budget was not tested. (vii) The October-2019 sessions were used only for development and could not serve as an external test set because their camera-orientation channels are constant within every session (a logging defect). (viii) The diagnosis of E-DDQN's weak primary result (Section V-G) rests on three random draws and one feature-set restriction: it shows that the draw matters, but feature sets and discount factors were still selected, not assigned, so it does not isolate the effect of any single component. (ix) For the agents with class-balanced replay, the CS-MLP with class-balanced minibatches and the standard DQN (Huber loss with a ± 1 reward), the untuned rule D0 is the 5:1 Bayes cut-off only nominally (Section III-D); the primary rule D1 and all ranking metrics are unaffected.

F. Implications

For cybersickness detection from session data, results should be reported with participant- or session-grouped splits, alongside trivial policies and an exposure-time baseline, because here no primary model ranked seconds significantly better than elapsed time alone. When neural or RL detectors are compared under a small search budget, the random-search draw should be varied or at least reported: here E-DDQN's within-fold ROC-AUC spanned 0.175 across three draws. For safety-prioritized detection, an asymmetric reward should be treated as what it is—a cost-sensitive operating point, which the class mix of replay or minibatches also shifts—and RL detectors should be compared with cost-weighted supervised models at matched recall or cost. Claims of robustness to manipulation should state which inputs the detector actually uses and evaluate decisions at fixed thresholds, since an unchanged ranking is not protection. Under grouped evaluation, participant-profile features deserve particular care, because with few groups they can fit group-level label rates that do not transfer.

VII. Conclusion

We asked whether an asymmetric-reward dueling Double DQN (E-DDQN) improves safety-prioritized cybersickness detection in VR. On the public UFFCSData gameplay dataset the answer is no. On logged data the detection problem is a contextual bandit, and the 5:1 asymmetric reward induces exactly the Bayes threshold of 5:1 cost-sensitive classification. Under a nested, session-grouped protocol fixed before testing, E-DDQN reached a mean cost over seeds of 0.663 and ROC-AUC 0.538, did not outperform a standard DQN (cost 0.584, ROC-AUC 0.661) and had a higher seed-ensemble cost than a cost-weighted random forest in every outer fold, marginally in one (cost 0.491, ROC-AUC 0.688; seed ensembles 0.726 vs. 0.485; Holm-adjusted $p = 0.006$ by group bootstrap and 0.083 by the more conservative exact sign-flip test). A re-tuning with two further random search draws, specified before it was run, found no draw in which E-DDQN had a significantly lower cost than the standard DQN, and each agent ranked seconds significantly better in one draw; it also showed that E-DDQN's weak primary result reflects its search draw: across draws its within-fold ROC-AUC ranged from 0.539 to 0.714 and its cost from 0.549 to 0.663, still above the random forest's as a point estimate. A post-hoc logistic regression on elapsed time alone reached ROC-AUC 0.704, no primary model ranked seconds significantly better, and no model combined high recall with a controlled false-alarm rate or was shown, with linkage groups as the unit of inference, to improve on always alarming at the 5:1 ratio. Session grouping was essential to these conclusions: splits that share sessions between training and test give near-perfect scores on these data, whereas session-grouped evaluation of the same features gives ROC-AUC 0.523. Bounded telemetry perturbations changed ROC-AUC by at most 0.089, which is not evidence of robustness: most selected configurations use exposure-type inputs, and an adaptive adversary still changed decisions at fixed thresholds. Progress on this problem is likely to require datasets with physiological or eye-tracking signals, verified participant identifiers and, for adversarial claims, manipulations applied to participants with measured effects, and evaluations that include grouped splits, trivial and exposure-time baselines, cost-weighted comparators and more than one random search draw per model.

Data and Code Availability

The data are the publicly available UFFCSData release [18] (AGPL-3.0). All code, the evaluation protocol (`configs/protocol.json`, committed before any model was trained on the evaluation sessions), the specification of the post-hoc analyses (`configs/posthoc.json`; which analyses were committed before being run and which were added later is stated in Supplementary Sec. S5), the split tables, the per-sample predictions of every model and the scripts that produce every number, table and figure of this paper are archived at [DOI to be added on deposit]. A single script regenerates everything from the raw XML files. Seeds are fixed for data splitting (20261012), hyper-parameter search (777; 801–802 for the post-hoc baselines; six configuration seeds between 9120 and 9202 for the search-draw analysis) and final training (1000–1004); CPU execution with one thread per process was used, and minor floating-point differences across hardware are possible.

Ethics Statement

This study is a secondary analysis of the publicly released UFFCSData dataset [18], whose session files contain no names or contact details of participants. No new data were collected from human participants, and no attempt was made to identify individuals; the session-linkage groups (Section III-B) are used only to keep sessions that may belong to the same person in

the same data partition. [Add your institution's ruling if it requires a formal exemption for secondary analysis of public data, and the ethics approval of the original data collection if the journal asks for it.]

Author Contributions

[To be completed by the authors with CRediT roles]: R. Darabee: [roles]; M. Muniandy: [roles]; O. Shindi: [roles]; C. W. Khuen: [roles]; E. H. Sumiea: [roles]. All authors read and approved the final manuscript.

Use of Generative AI

The authors used Claude, an AI assistant developed by Anthropic ([version and period of use]), to help re-implement and run the analysis pipeline (data parsing, evaluation protocol, statistical analysis and figures), to check numerical consistency and bibliographic references, and to draft and edit the manuscript text. The authors reviewed and verified the code, results and text and take full responsibility for the content of this article. [Confirm that this describes your use, and adapt it to the target journal's policy.]

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Conflict of Interest

The authors declare no conflict of interest.

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References

[1] P. Caserman, A. Garcia-Agundez, A. Gámez Zerban, and S. Göbel, "Cybersickness in current-generation virtual reality head-mounted displays: Systematic review and outlook," *Virtual Reality*, vol. 25, no. 4, pp. 1153–1170, 2021.

[2] W.-J. Tseng, E. Bonnail, M. McGill, M. Khamis, E. Lecolinet, S. Huron, and J. Gugenheimer, "The dark side of perceptual manipulations in virtual reality," in *Proc. CHI Conf. Human Factors in Computing Systems*, 2022.

[3] B. Odeleye, G. Loukas, R. Heartfield, and F. Spyridonis, "Detecting framerate-oriented cyber attacks on user experience in virtual reality," in *Proc. 1st Int. Workshop Security for XR and XR for Security (VR4Sec), held with SOUPS 2021*, 2021, pp. 1–5.

[4] R. K. Kundu, M. Denton, G. Mongalo, P. Calyam, and K. A. Hoque, "Securing virtual reality experiences: Unveiling and tackling cybersickness attacks with explainable AI," *IEEE Trans. Dependable Secure Comput.*, vol. 22, no. 6, pp. 6040–6057, 2025, arXiv:2503.13419.

[5] M. S. Dennison, A. Z. Wisti, and M. D'Zmura, "Use of physiological signals to predict cybersickness," *Displays*, vol. 44, pp. 42–52, 2016.

[6] R. Islam, Y. Lee, M. Jaloli, I. Muhammad, D. Zhu, P. Rad, Y. Huang, and J. Quarles, "Automatic detection and prediction of cybersickness severity using deep neural networks from user's physiological signals," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR)*, 2020, pp. 400–411.

[7] R. Islam, K. Desai, and J. Quarles, "Cybersickness prediction from integrated HMD's sensors: A multimodal deep fusion approach using eye-tracking and head-tracking data," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR)*, 2021, pp. 31–40, arXiv:2108.06437.

[8] M. Liu, B. Yang, M. Xu, P. Zan, L. Chen, and X. Xia, "Exploring quantitative assessment of cybersickness in virtual reality using EEG signals and a CNN-ECA-LSTM network," *Displays*, vol. 81, p. 102602, 2024.

[9] M. Yalcin, A. Halbig, M. Fischbach, and M. E. Latoschik, "Automatic cybersickness detection by deep learning of augmented physiological data from off-the-shelf consumer-grade sensors," *Frontiers in Virtual Reality*, vol. 5, p. 1364207, 2024.

[10] D. Jeong and K. Han, "PRECYSE: Predicting cybersickness using transformer for multimodal time-series sensor data," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 8, no. 2, pp. 42:1–42:24, 2024.

[11] E. Lin, Q. Chen, and X. Qi, "Deep reinforcement learning for imbalanced classification," *Applied Intelligence*, vol. 50, pp. 2488–2502, 2020.

[12] J. Yang, R. El-Bouri, O. O'Donoghue, A. S. Lachapelle, A. A. S. Soltan, D. W. Eyre, L. Lu, and D. A. Clifton, "Deep reinforcement learning for multi-class imbalanced training: Applications in healthcare," *Mach. Learn.*, vol. 113, pp. 2655–2674, 2024.

[13] H. van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *Proc. AAAI Conf. Artificial Intelligence*, 2016, pp. 2094–2100.

[14] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, and N. de Freitas, "Dueling network architectures for deep reinforcement learning," in *Proc. Int. Conf. Machine Learning (ICML)*, 2016, pp. 1995–2003.

[15] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.

[16] R. K. Kundu, R. Islam, P. Calyam, and K. A. Hoque, "TruVR: Trustworthy cybersickness detection using explainable machine learning," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality (ISMAR)*, 2022, pp. 777–786.

[17] C. Elkan, "The foundations of cost-sensitive learning," in *Proc. Int. Joint Conf. Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.

[18] T. Porcino *et al.*, "UFFCSData: Raw and selected datasets with user tests reporting the level of discomfort during virtual experiments," https://github.com/tmp1986/UFFCSData, 2019, AGPL-3.0.

[19] M. Dudík, D. Hsu, S. Kale, N. Karampatziakis, J. Langford, L. Reyzin, and T. Zhang, "Efficient optimal learning for contextual bandits," in *Proc. Conf. Uncertainty in Artificial Intelligence (UAI)*, 2011, arXiv:1106.2369.

[20] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Trans. Knowl. Discov. Data*, vol. 6, no. 4, p. 15, 2012.

[21] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal, "Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness," *Int. J. Aviation Psychology*, vol. 3, no. 3, pp. 203–220, 1993.

[22] R. K. Kundu, R. Islam, J. Quarles, and K. A. Hoque, "LiteVR: Interpretable and lightweight cybersickness detection using explainable AI," in *Proc. IEEE Conf. Virtual Reality and 3D User Interfaces (VR)*, 2023, pp. 609–619.

[23] R. K. Kundu, O. Y. Elsaid, P. Calyam, and K. A. Hoque, "VR-LENS: Super learning-based cybersickness detection and explainable AI-guided deployment in virtual reality," in *Proc. 28th Int. Conf. Intelligent User Interfaces (IUI)*, 2023, pp. 819–834.

[24] A. H. X. Yang, N. Kasabov, and Y. O. Cakmak, "Machine learning methods for the study of cybersickness: A systematic review," *Brain Informatics*, vol. 9, p. 24, 2022.

[25] T. Porcino, D. Trevisan, and E. Clua, "An experimental methodology to capture user and gameplay data tied to cybersickness," in *Proc. Workshop on Extended Reality in Games (XR in Games), held with the ACM Int. Conf. Interactive Media Experiences (IMX)*, 2021.

[26] T. Porcino, E. O. Rodrigues, F. Bernardini, D. Trevisan, and E. Clua, "Identifying cybersickness causes in virtual reality games using symbolic machine learning algorithms," *Entertainment Computing*, vol. 41, 2022, art. no. 100473.

[27] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[28] B. Odeleye, G. Loukas, R. Heartfield, G. Sakellari, E. Panaousis, and F. Spyridonis, "Virtually secure: A taxonomic assessment of cybersecurity challenges in virtual reality environments," *Computers & Security*, vol. 124, p. 102951, 2023.

[29] A. Giaretta, "Security and privacy in virtual reality: A literature survey," *Virtual Reality*, vol. 29, p. 10, 2025.

[30] K. Cheng, J. F. Tian, T. Kohno, and F. Roesner, "Exploring user reactions and mental models towards perceptual manipulation

attacks in mixed reality," in *Proc. 32nd USENIX Security Symp.* USENIX Association, 2023. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/cheng-kaiming

[31] I. Ahmed, R. K. Kundu, and K. A. Hoque, "Adversarial VR: An open-source testbed for evaluating adversarial robustness of VR cybersickness detection and mitigation," in *Proc. IEEE Int. Symp. Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2025.

[32] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[33] J. García and F. Fernández, "A comprehensive survey on safe reinforcement learning," *J. Machine Learning Research*, vol. 16, no. 42, pp. 1437–1480, 2015.

[34] T. Porcino, D. Trevisan, and E. Clua, "DEMO: Using gameplay data to classify cybersickness level in virtual environments," in *Anais Estendidos do Simpósio de Realidade Virtual e Aumentada (SVR)*. SBC, 2019. [Online]. Available: https://sol.sbc.org.br/index.php/svr_estendido/article/view/8462

[35] T. Lattimore and C. Szepesvári, *Bandit Algorithms*. Cambridge University Press, 2020.

[36] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Machine Learning (ICML)*, 2015, pp. 448–456.

[37] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.

[38] P. J. Huber, "Robust estimation of a location parameter," *Ann. Math. Statist.*, vol. 35, no. 1, pp. 73–101, 1964.

[39] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2015, pp. 1440–1448.

[40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.

[41] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.

[42] T. P. Lillicrap *et al.*, "Continuous control with deep reinforcement learning," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2016.

[43] L.-J. Lin, "Self-improving reactive agents based on reinforcement learning, planning and teaching," *Machine Learning*, vol. 8, no. 3–4, pp. 293–321, 1992.

[44] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[45] R. S. Kennedy, K. M. Stanney, and W. P. Dunlap, "Duration and exposure to virtual environments: Sickness curves during and across sessions," *Presence: Teleoperators and Virtual Environments*, vol. 9, no. 5, pp. 463–472, 2000.

[46] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, p. 6, 2020.

[47] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[48] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Bootstrap-based improvements for inference with clustered errors," *Review of Economics and Statistics*, vol. 90, no. 3, pp. 414–427, 2008.

[49] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[50] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.

[51] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

[52] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

[53] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 3145–3153.

[54] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[55] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, p. 91, 2006.